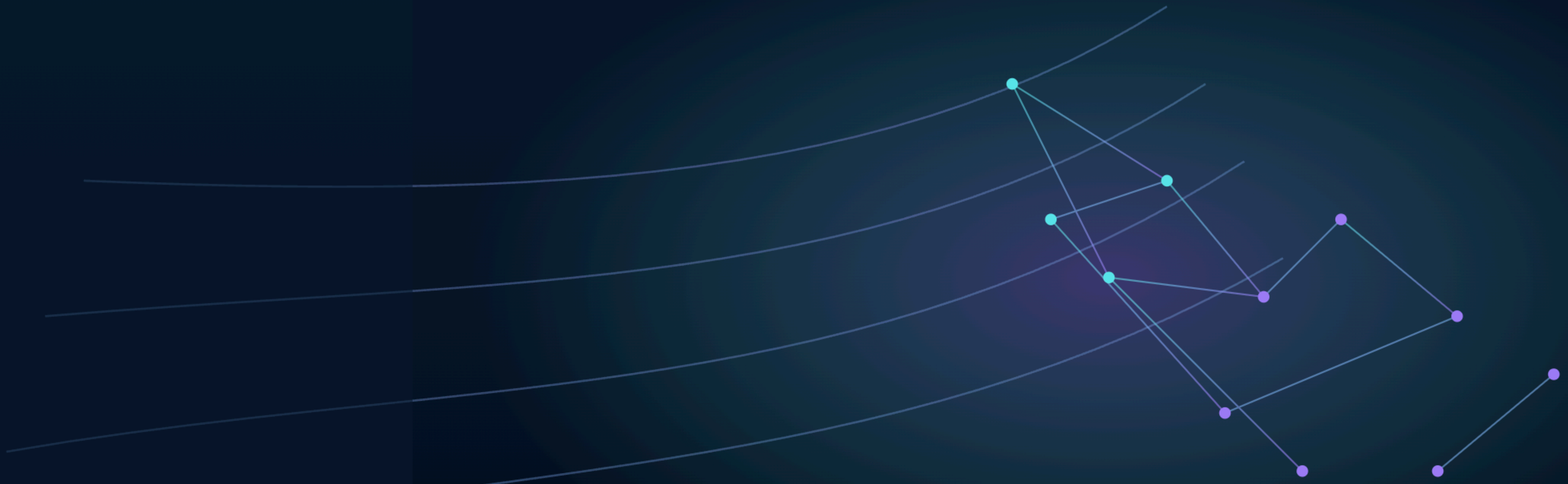
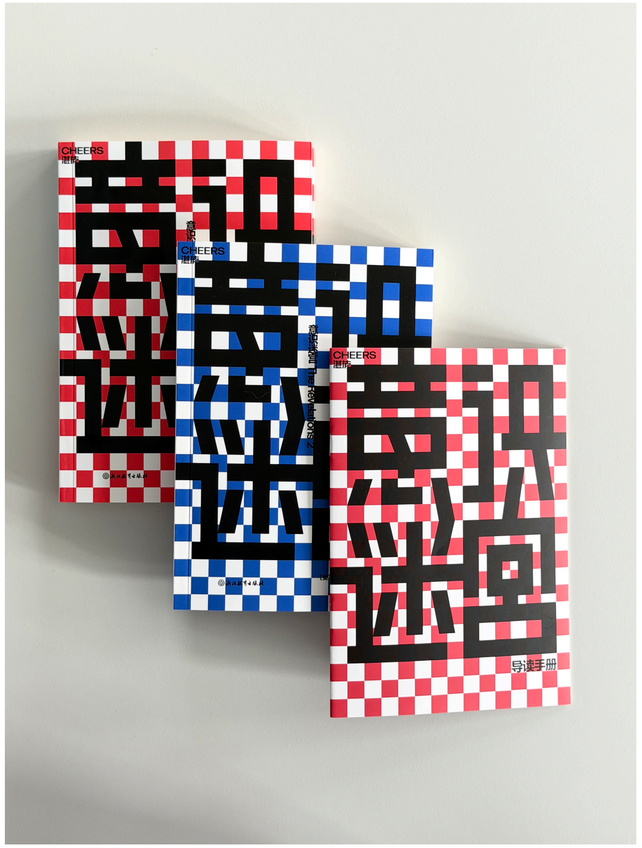
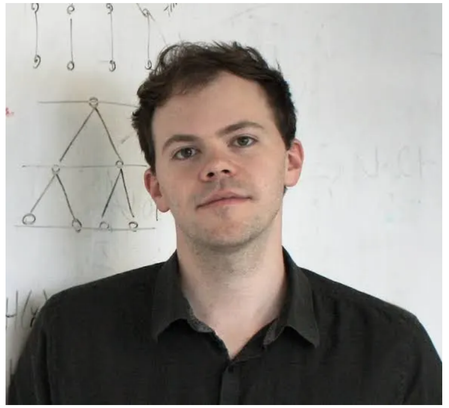
陶如意 2025-12-19
北京师范大学 系统科学学院



# 整体大于部分之和?

## 从复杂科学、涌现到意识



Erik Hoel

神经科学家

科幻小说作家

福布斯30位30岁以下科学家奖

因果涌现理论提出者

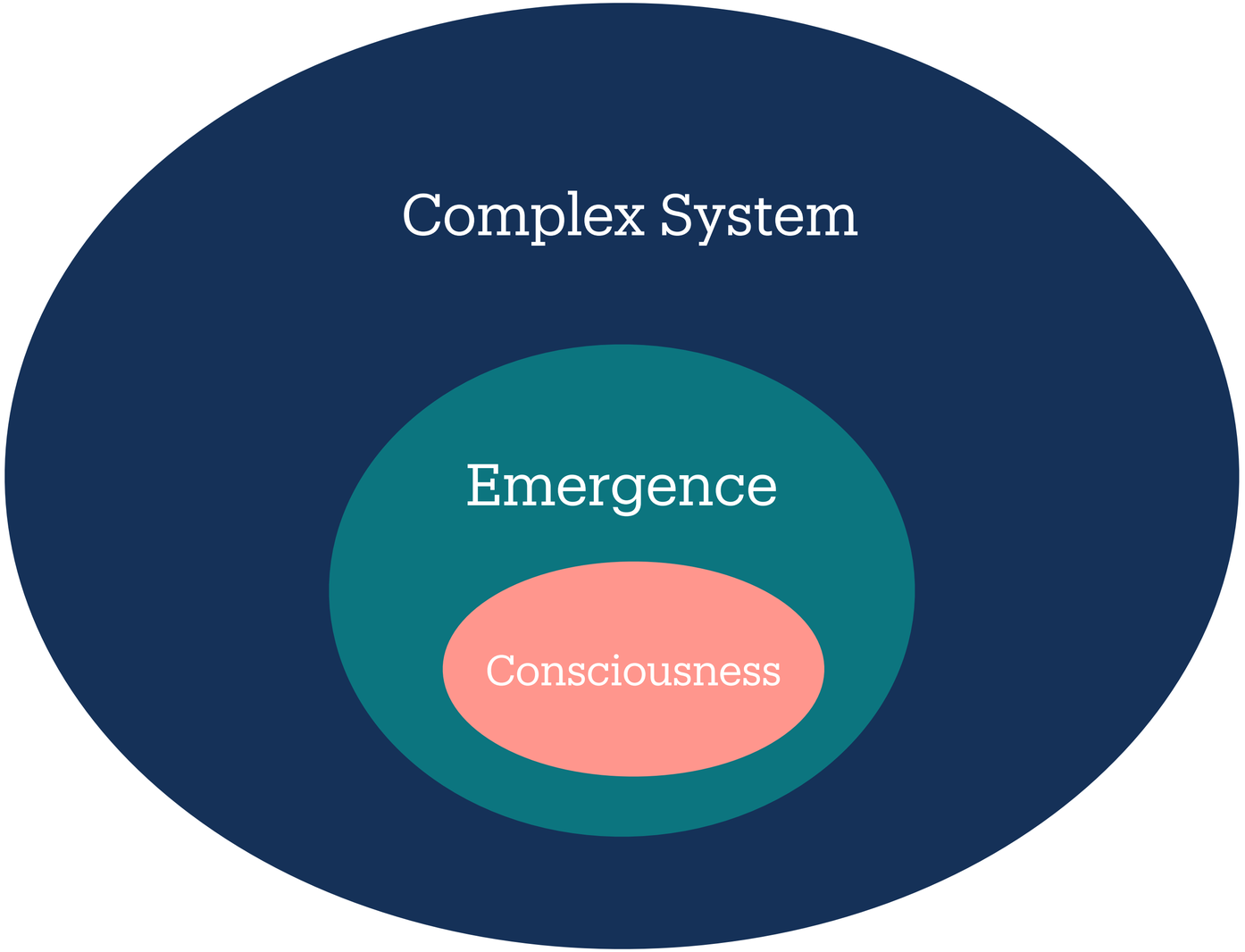
IIT理论贡献者

集智俱乐部

- 一部以意识科学研究领域为背景的谋杀悬疑小说。
- 这部兼具硬核科学底色与文学诗性的小说，以神经科学界的“圣杯”——意识本质研究为舞台，编织了一场横跨认知科学、存在主义哲学与技术伦理的多维狂想

# Outline

- **Complex Science & Complex System**（15min）
- **Emergence**（45min）
  - Modeling from bottom to up
  - Measuring from up to bottom
- **Consciousness**（45min）
  - Evidences in neural science
  - Theory of Consciousness



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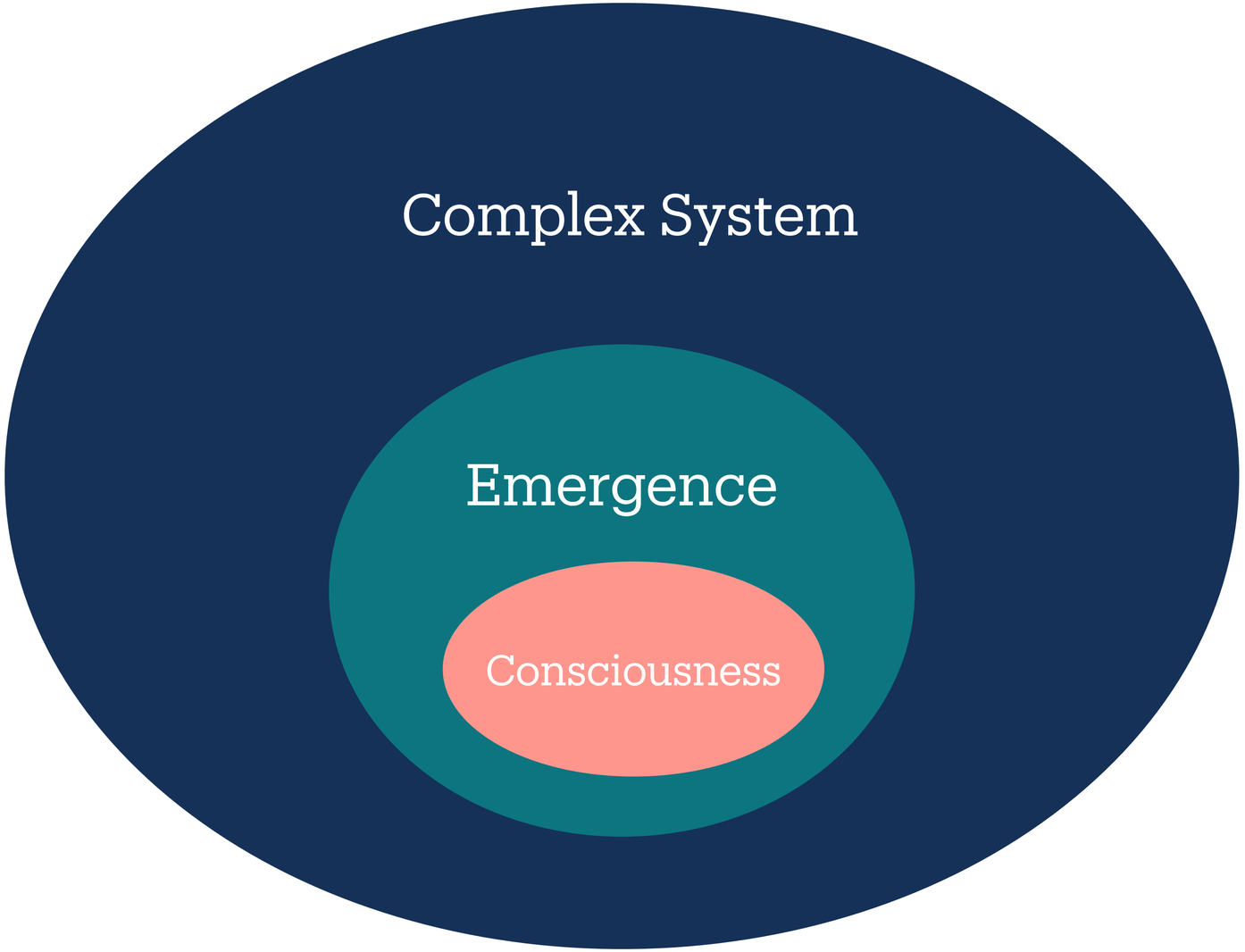
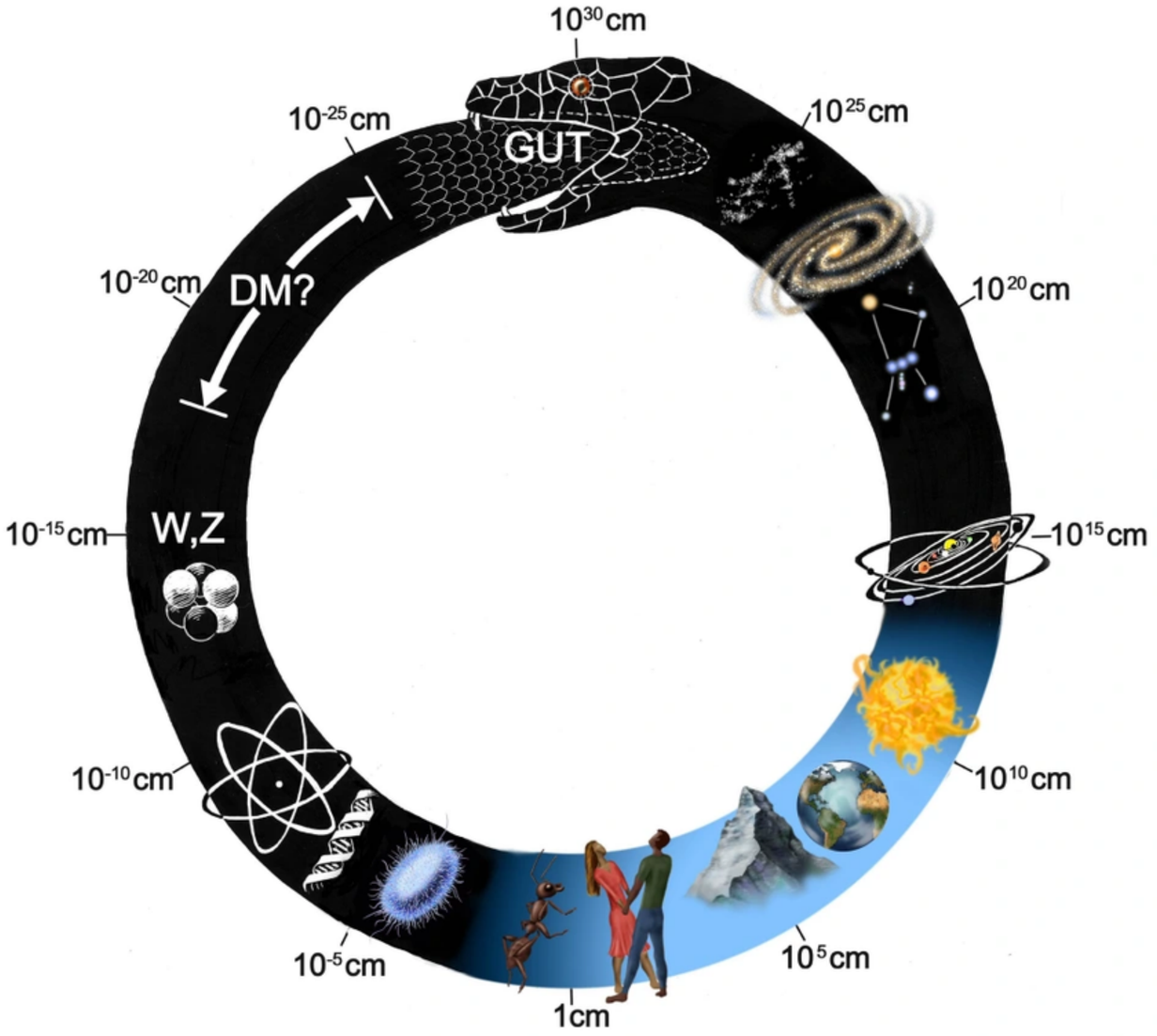


# Complex System

自20世纪以来，人类对于宇观层次和微观层次的对象已经有了较为深刻的科学认识

- 宇观层次
  - 宇宙大爆炸理论
  - 2016年证实引力波的存在

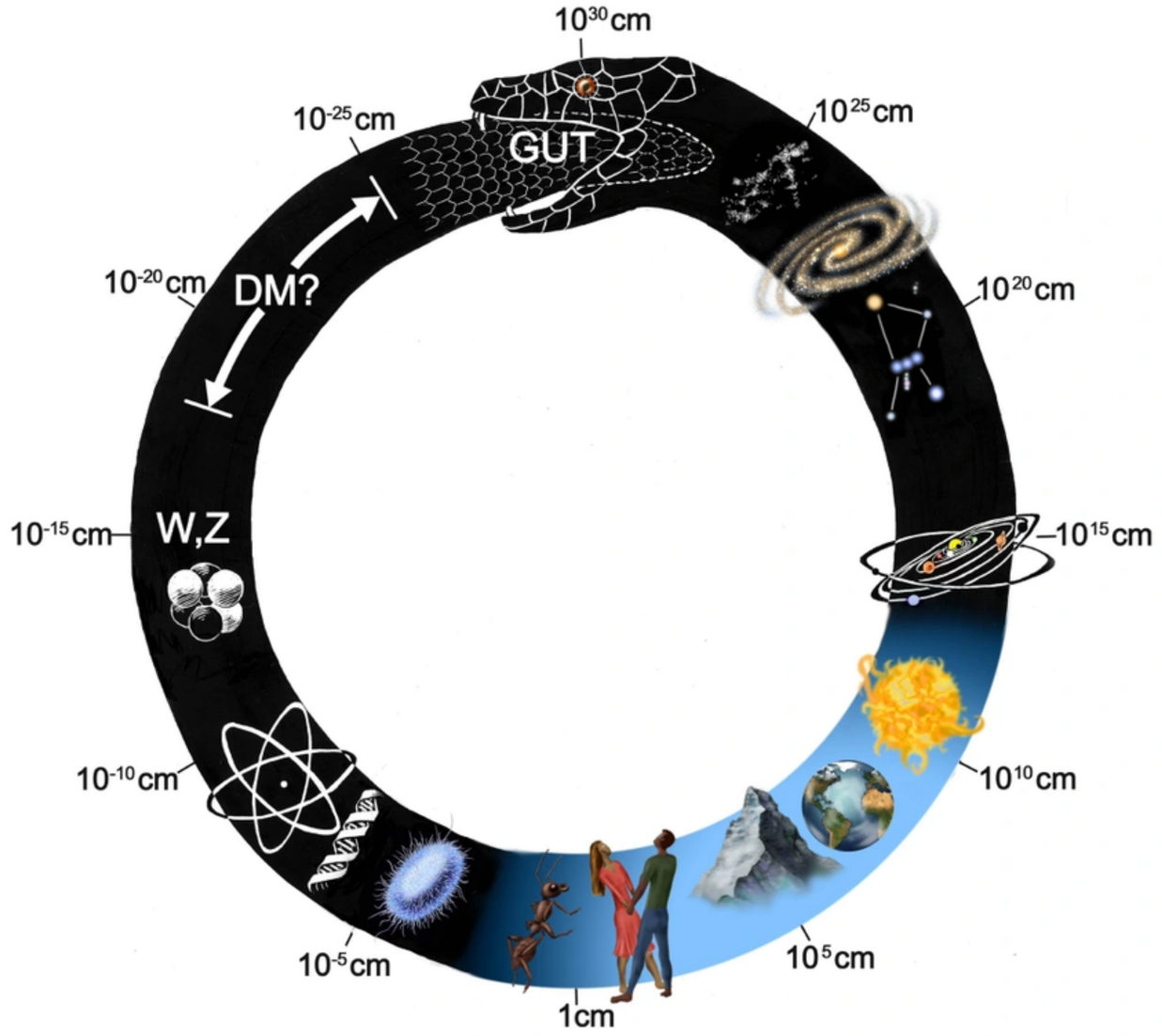
- 微观层次
  - 宇宙中目前已知的物质，都是有6种轻子和6中夸克组成
  - 2013年验证Higgs粒子的存在



# Complex System

但对于宏观尺度，许多问题反倒缺乏足够的科学认识

- 生命的起源?
- 为什么大脑会思考?
- 为什么我一买股票就跌?
- 气候变暖如何遏制?
- 气候与经济如何互相影响?
- 金融、社交网络、集群行为、交通控制、公司经营….




大量元素、复杂的相互作用　　元素属性 → 联系

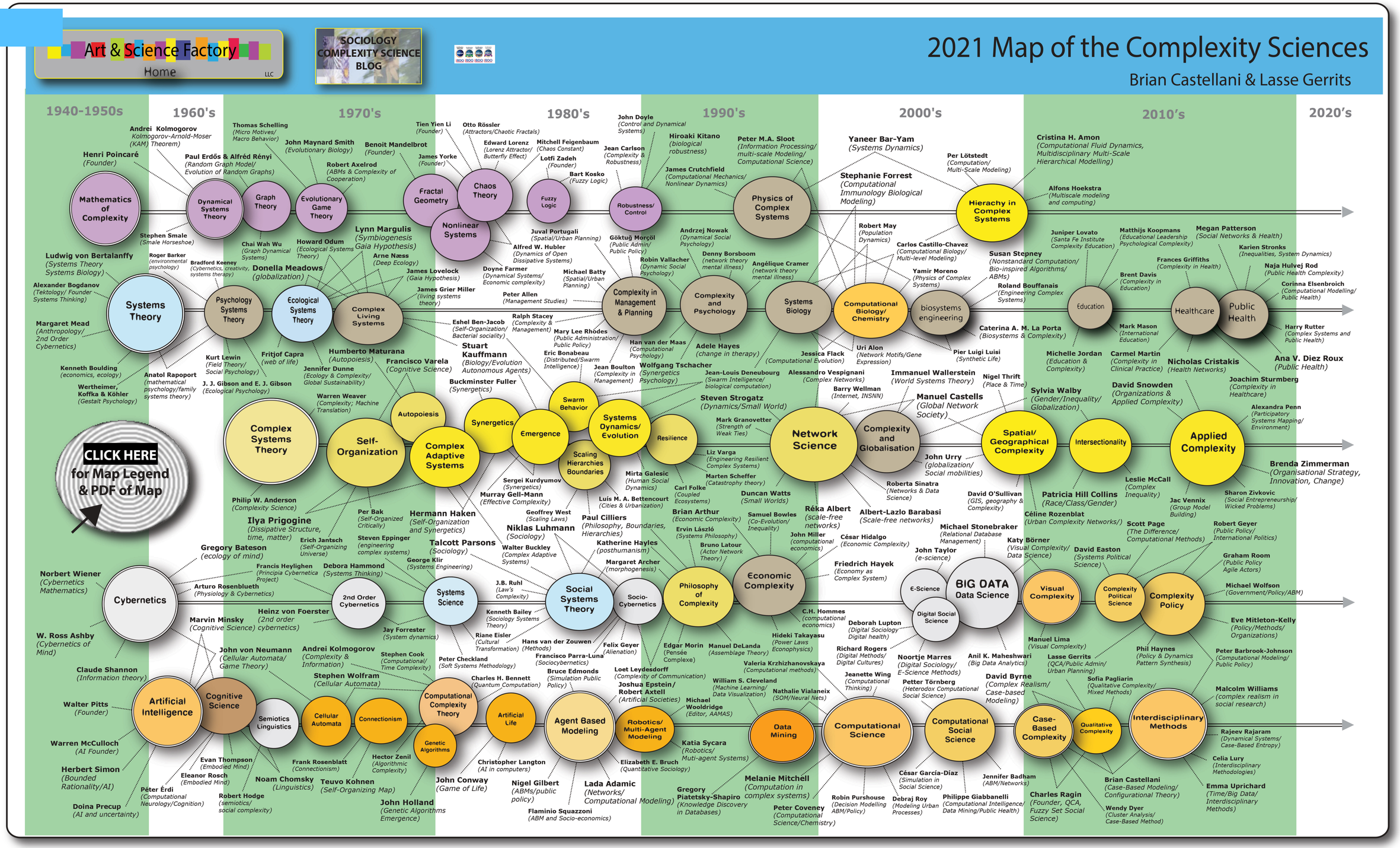
# Complex System

系统学的建立，实际上是一次科学革命，它的重要性绝不亚于相对论或者量子力学

—钱学森
1986.01.07

I Think the Next Century Will Be the Century of Complexity

—Stephen Hawking
January 23, 2000, San Jose Mercury News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SOCIOLOGY COMPLEXITY SCIENCE BLOG
2021 Map of the Complexity Sciences
Brian Castellani & Lasse Gerrits
1940-1950s
1960's
1970's
1980's
1990's
2000's
2010's
2020's
CLICK HERE
for Map Legend
& PDF of Map
Mathematics of Complexity
Dynamical Systems Theory
Graph Theory
Evolutionary Game Theory
Fractal Geometry
Chaos Theory
Nonlinear Systems
Fuzzy Logic
Robustness/ Control
Physics of Complex Systems
Hierachy in Complex Systems
Systems Theory
Psychology Systems Theory
Ecological Systems Theory
Complex Living Systems
Complexity in Management & Planning
Complexity and Psychology
Systems Biology
Computational Biology/ Chemistry
biosystems engineering
Education
Healthcare
Public Health
Complex Systems Theory
Self-Organization
Autopoiesis
Complex Adaptive Systems
Synergetics
Emergence
Swarm Behavior
Systems Dynamics/ Evolution
Scaling Hierarchies Boundaries
Resilience
Network Science
Complexity and Globalisation
Spatial/ Geographical Complexity
Intersectionality
Applied Complexity
Cybernetics
2nd Order Cybernetics
Systems Science
Social Systems Theory
Socio-Cybernetics
Philosophy of Complexity
Economic Complexity
E-Science
Digital Social Science
BIG DATA Data Science
Visual Complexity
Complexity Political Science
Complexity Policy
Artificial Intelligence
Cognitive Science
Semiotics Linguistics
Cellular Automata
Connectionism
Computational Complexity Theory
Genetic Algorithms
Artificial Life
Agent Based Modeling
Robotics/ Multi-Agent Modeling
Data Mining
Computational Science
Computational Social Science
Case-Based Complexity
Qualitative Complexity
Interdisciplinary Methods
Andrei Kolmogorov (Kolmogorov-Arnold-Moser (KAM) Theorem)
Henri Poincaré (Founder)
Paul Erdős & Alfréd Rényi (Random Graph Model/ Evolution of Random Graphs)
Thomas Schelling (Micro Motives/ Macro Behavior)
John Maynard Smith (Evolutionary Biology)
Robert Axelrod (ABMs & Complexity of Cooperation)
Benoit Mandelbrot (Founder)
Tien Yien Li (Founder)
Otto Rössler (Attractors/Chaotic Fractals)
James Yorke (Founder)
Edward Lorenz (Lorenz Attractor/ Butterfly Effect)
Mitchell Feigenbaum (Chaos Constant)
Lotfi Zadeh (Founder)
Bart Kosko (Fuzzy Logic)
John Doyle (Control and Dynamical Systems)
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Peter M.A. Sloot (Information Processing/ multi-scale Modeling/ Computational Science)
Yaneer Bar-Yam (Systems Dynamics)
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Per Lötstedt (Computation/ Multi-Scale Modeling)
Cristina H. Amon (Computational Fluid Dynamics, Multidisciplinary Multi-Scale Hierarchical Modelling)
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Carmel Martin (Complexity in Clinical Practice)
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Margaret Archer (morphogenesis)
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Albert-Lazlo Barabasi (Scale-free networks)
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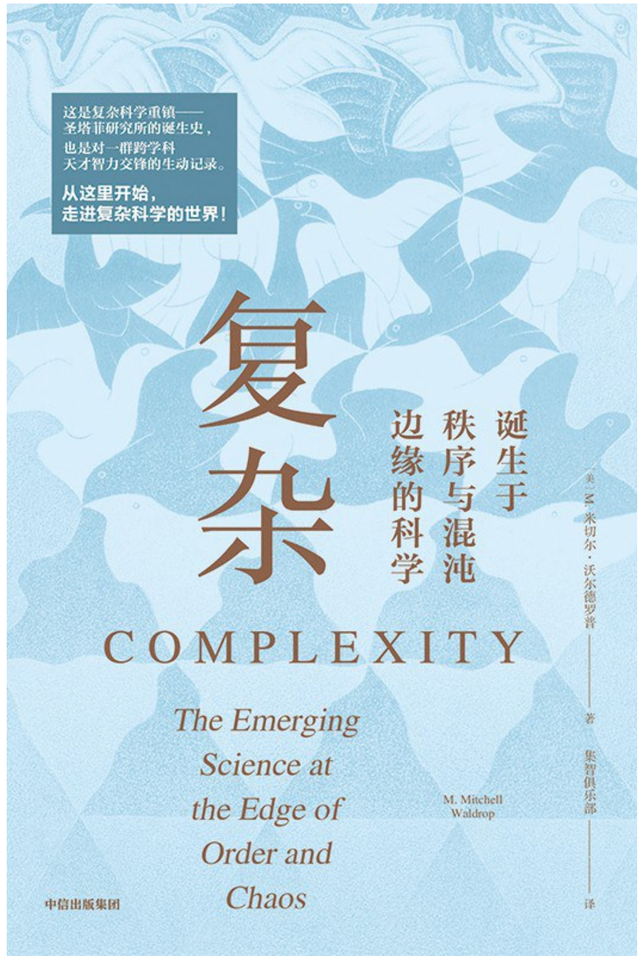
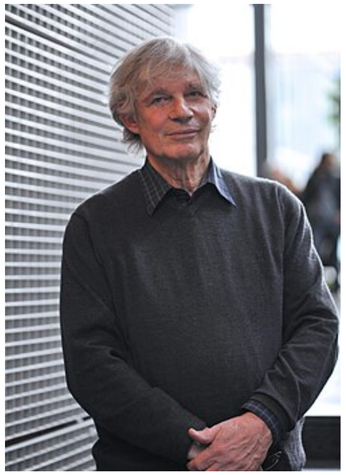
Wendy Dyer (Cluster Analysis/ Case-Based Method)

# Complex System

复杂科学如何创立的故事

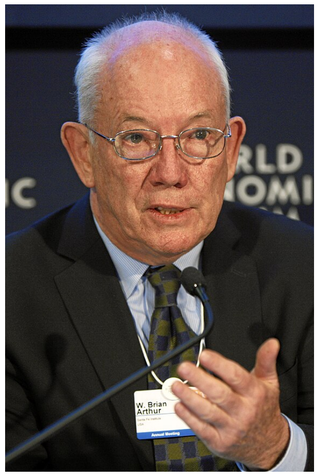



圣塔菲研究所



斯图亚特 考夫曼

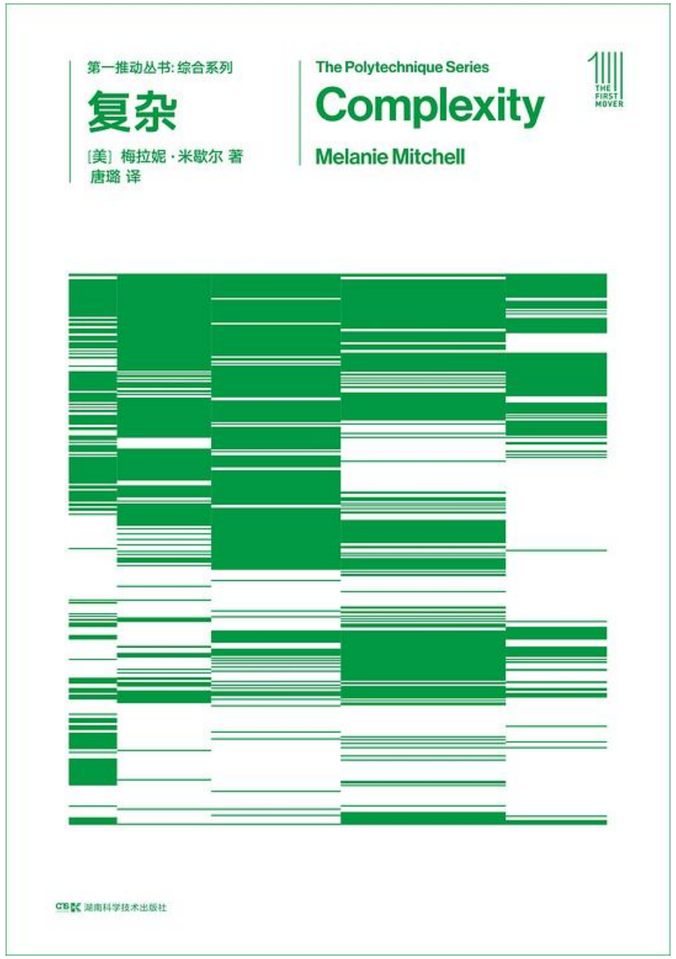
自催化理论




布莱恩 阿瑟

报酬递增理论

复杂科学基本概念与方法



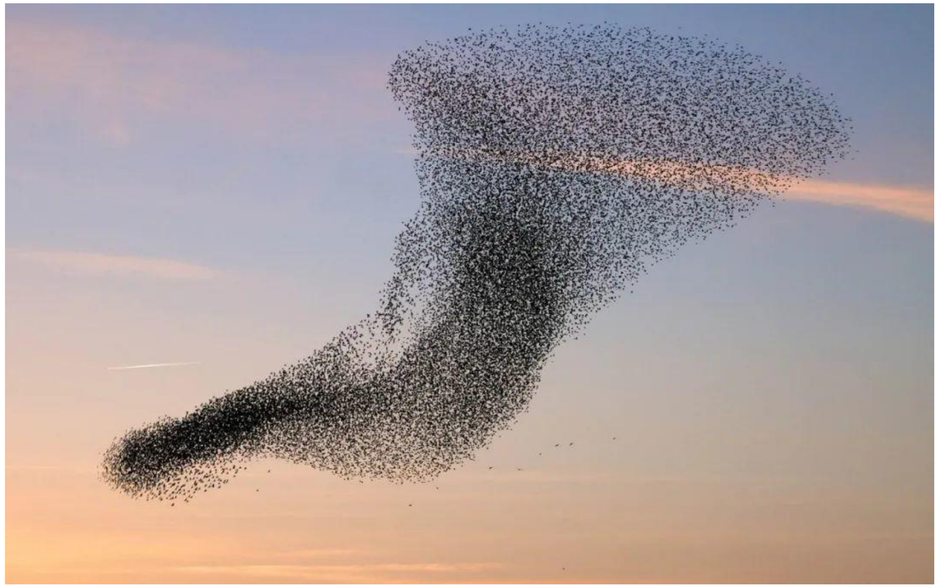

- 分形
- 相变与临界
- 自组织
- …
- 复杂系统的稳定性与韧性
- 涌现
- 混沌

**复杂系统的普适属性**

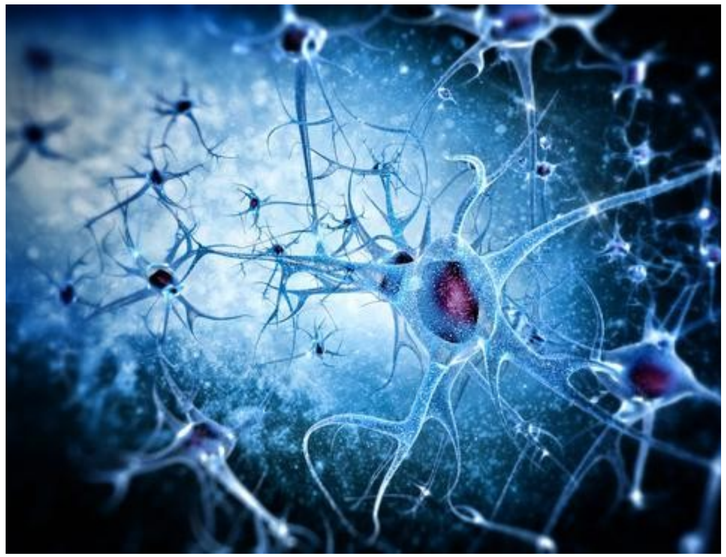
**复杂世界 vs 简单规则**

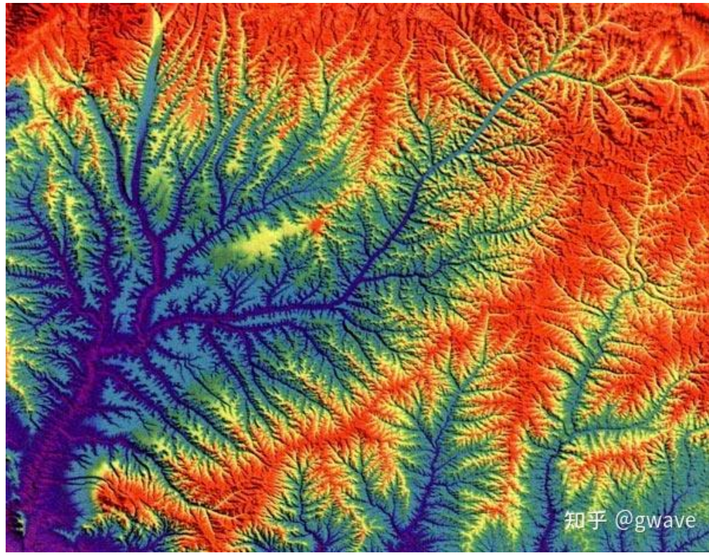
- Scaling Theory
- 图形生长
- 涌现计算
- 混沌边缘
- 非线性动力学
- 网络
- …

# Emergence



涌现是复杂系统的核心特征





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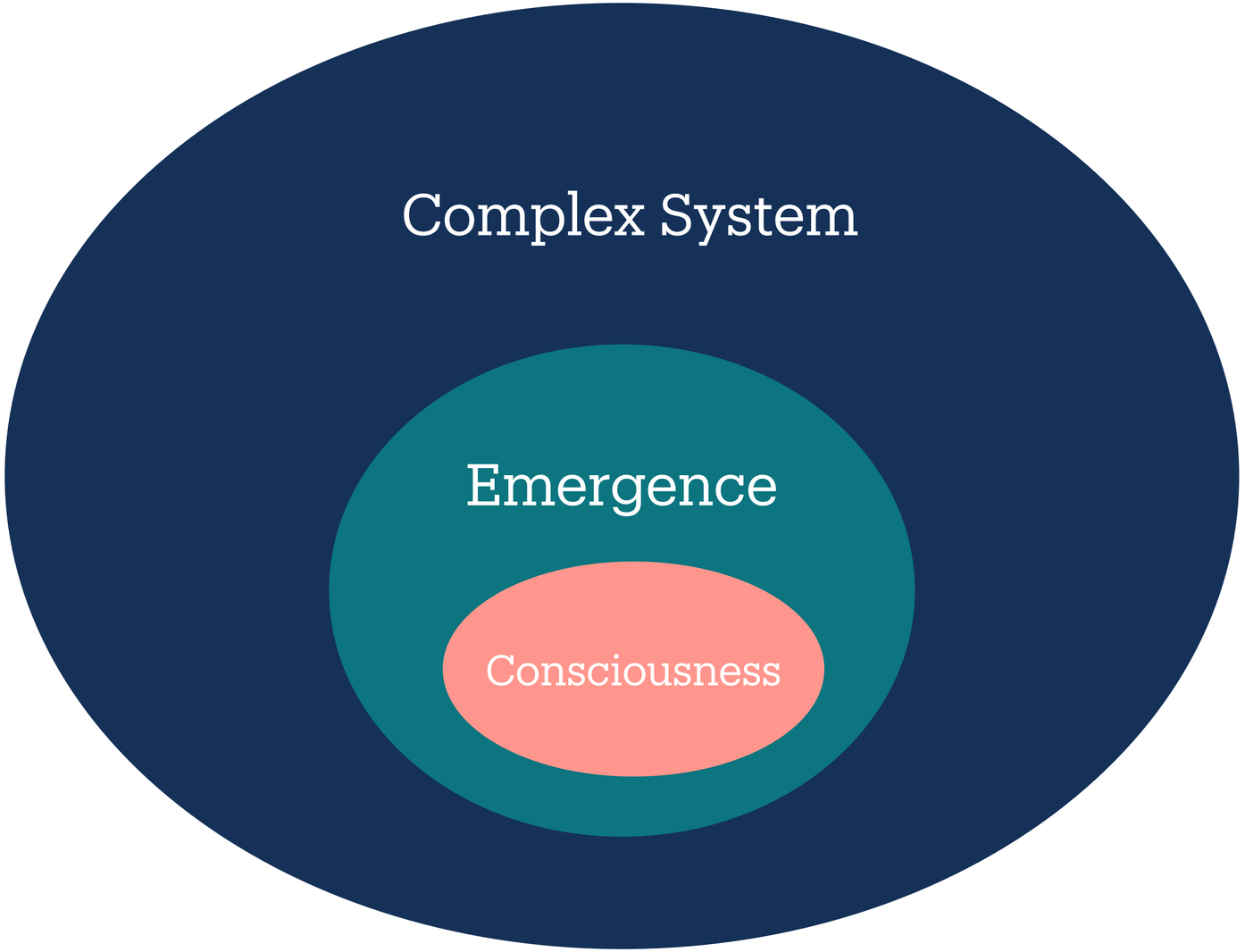


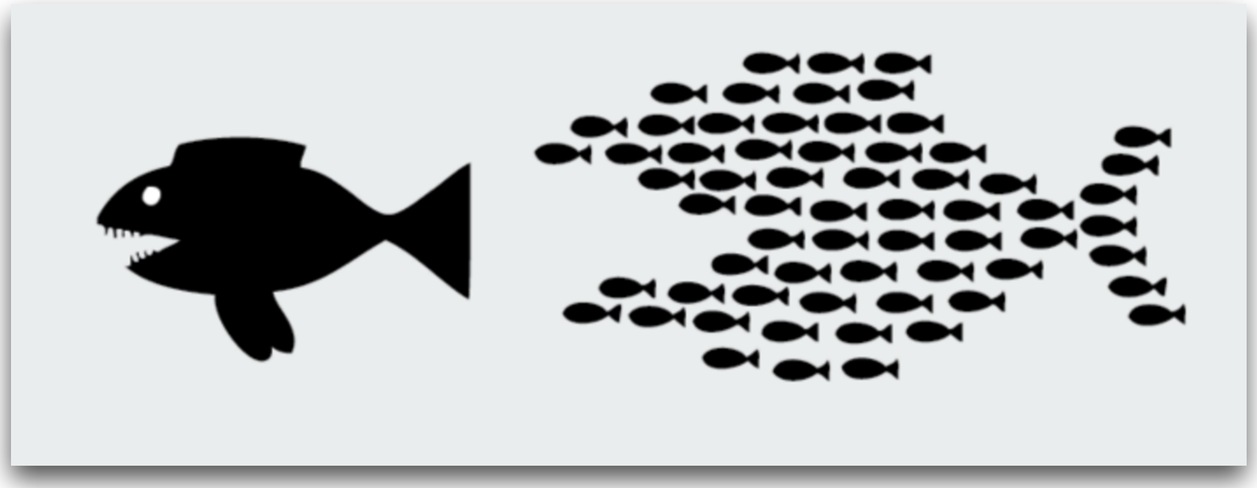
# What is Emergence

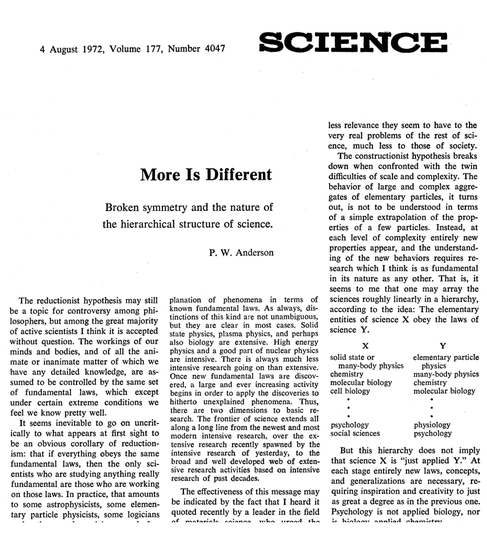
整体大于部分之和

— 亚里士多德

More is different

— Philip W. Anderson





4 August 1972, Volume 177, Number 4047 **SCIENCE**

**More Is Different**

Broken symmetry and the nature of the hierarchical structure of science.

P. W. Anderson

The reductionist hypothesis may still be a topic for controversy among philosophers, but among the great majority of active scientists I think it is accepted without question. The workings of our minds and bodies, and of all the animate or inanimate matter of which we have any detailed knowledge, are assumed to be controlled by the same set of fundamental laws, which except under certain extreme conditions we feel we know pretty well.

It seems inevitable to go on uncritically to what appears at first sight to be an obvious corollary of reductionism: that if everything obeys the same fundamental laws, then the only scientists who are studying anything really fundamental are those who are working on those laws. In practice, that amounts to some astrophysicists, some elementary particle physicists, some logicians

planation of phenomena in terms of known fundamental laws. As always, distinctions of this kind are not unambiguous, but they are clear in most cases. Solid state physics, plasma physics, and perhaps also biology are extensive. High energy physics and a good part of nuclear physics are intensive. There is always much less intensive research going on than extensive. Once new fundamental laws are discovered, a large and ever increasing activity begins in order to apply the discoveries to hitherto unexplained phenomena. Thus, there are two dimensions to basic research. The frontier of science extends all along a long line from the newest and most modern intensive research, over the extensive research recently spawned by the intensive research of yesterday, to the broad and well developed web of extensive research activities based on intensive research of past decades.

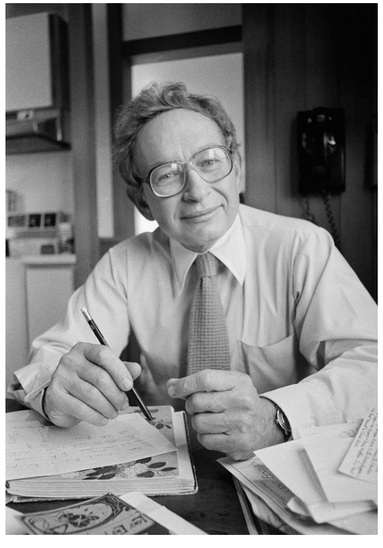
The effectiveness of this message may be indicated by the fact that I heard it quoted recently by a leader in the field

less relevance they seem to have to the very real problems of the rest of science, much less to those of society.

The constructionist hypothesis breaks down when confronted with the twin difficulties of scale and complexity. The behavior of large and complex aggregates of elementary particles, it turns out, is not to be understood in terms of a simple extrapolation of the properties of a few particles. Instead, at each level of complexity entirely new properties appear, and the understanding of the new behaviors requires research which I think is as fundamental in its nature as any other. That is, it seems to me that one may array the sciences roughly linearly in a hierarchy, according to the idea: The elementary entities of science X obey the laws of science Y.

| X | Y |
|---|---|
| solid state or many-body physics | elementary particle physics |
| chemistry | many-body physics |
| molecular biology | chemistry |
| cell biology | molecular biology |
| • | • |
| • | • |
| • | • |
| psychology | physiology |
| social sciences | psychology |

But this hierarchy does not imply that science X is "just applied Y." At each stage entirely new laws, concepts, and generalizations are necessary, requiring inspiration and creativity to just as great a degree as in the previous one. Psychology is not applied biology, nor

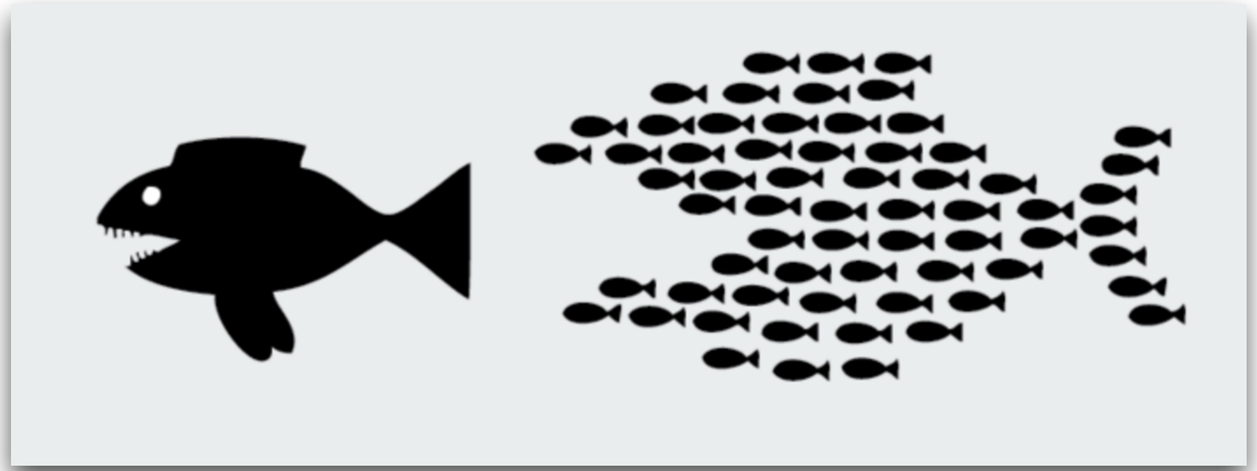


# What is Emergence

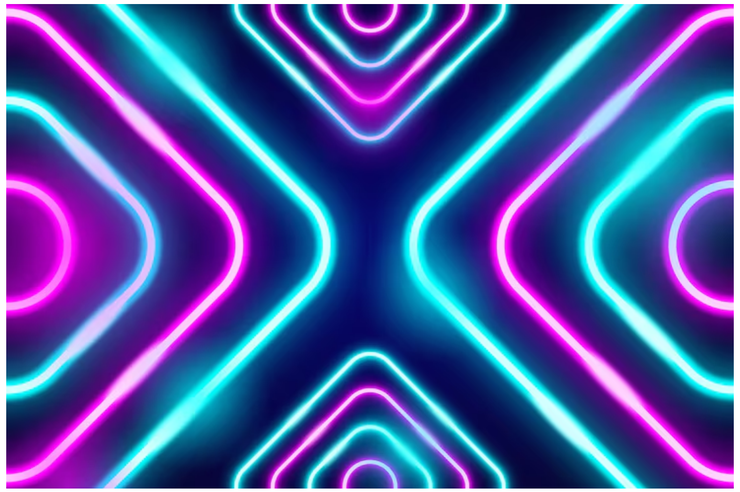
In a process of emergence, we observe something (for instance the appearance of order or organization) and ask how this is possible, since we assume causality: every effect should have a cause.

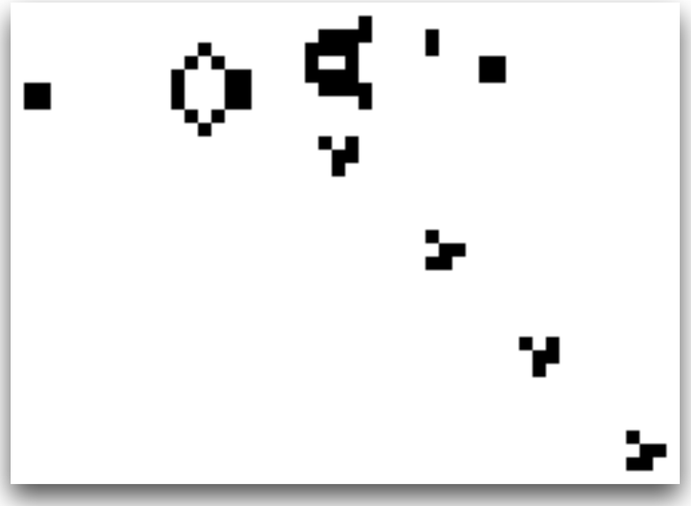
The surprising aspect in a process of emergence is the observation of an effect without an apparent cause

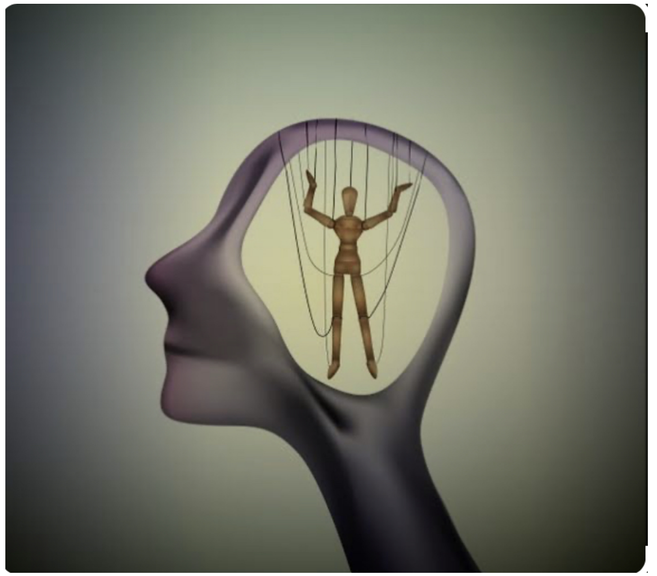
—Jochen Fromm



# 涌现的分类







Nominal Emergence

静态的

Weak Emergence

原则上被计算机模拟

Strong Emergence:

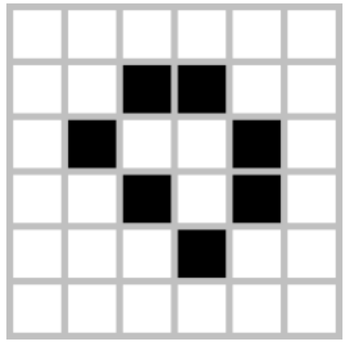
无法被计算机模拟

向下因果

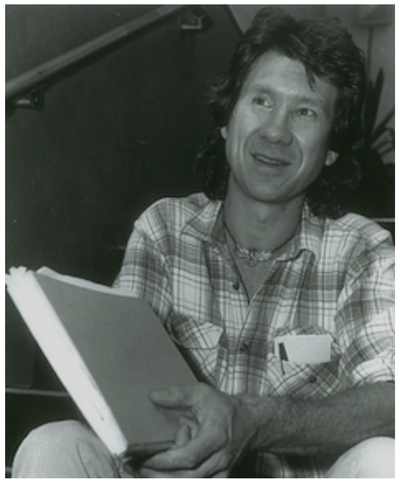
Bedau M. Downward causation and the autonomy of weak emergence[J]. Principia: an international journal of epistemology, 2002, 6(1): 5-50.

# 涌现的分类

生命游戏



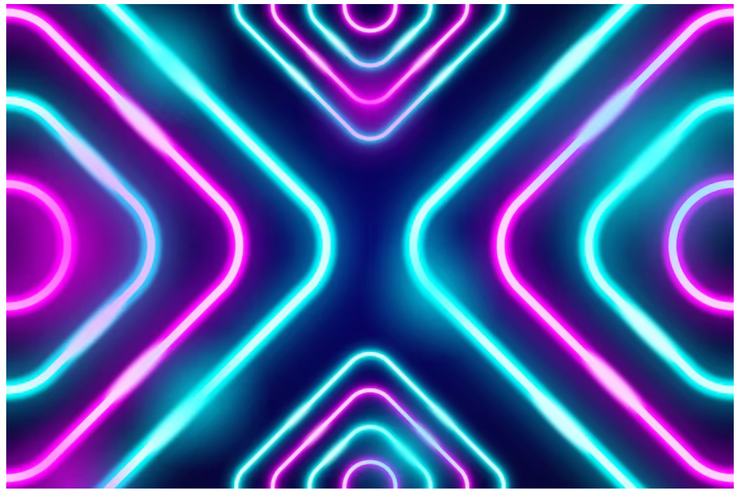
Conway

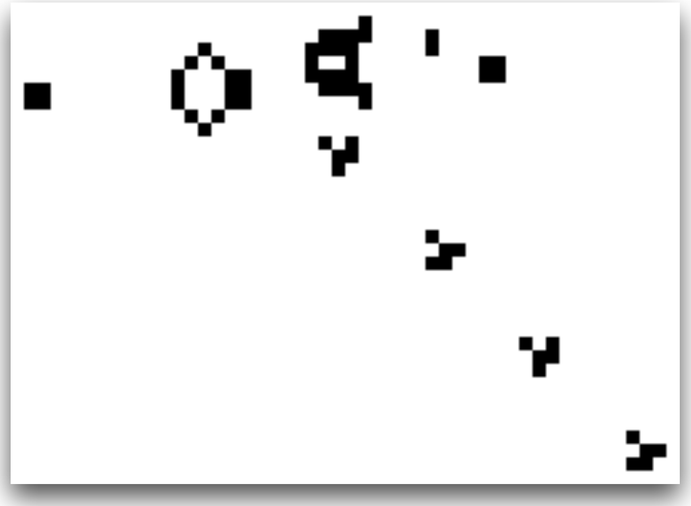


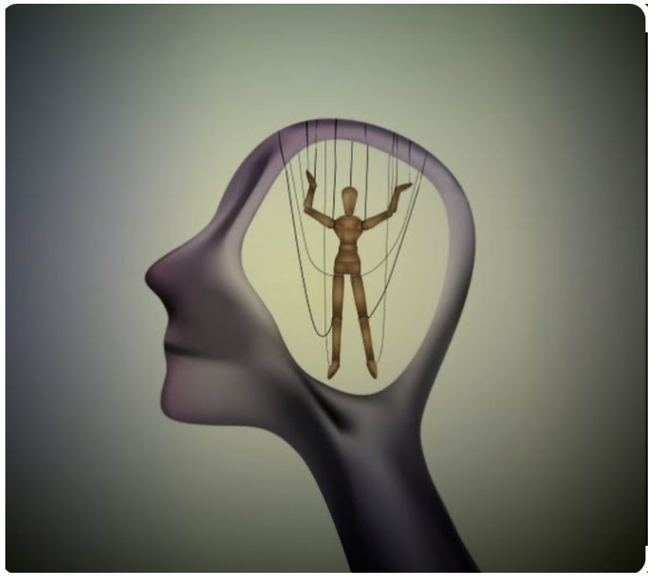
Langton

人工生命

# 涌现的分类







Nominal Emergence

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# Emergence

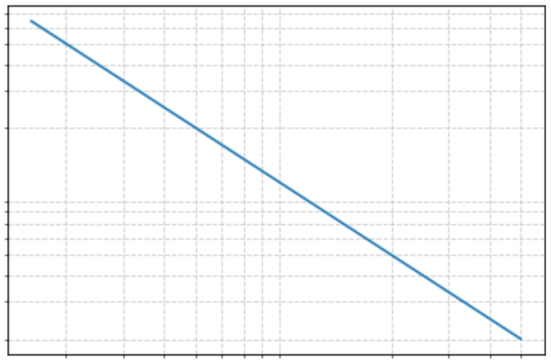
- 自下而上的建模
  - 复杂世界，简单规则

- 自上而下的计算
  - 从现象出发，研究涌现的量化

# An example: Fractal

- 分形来自于对欧式几何维数的拓展，应用于对现实世界的测度
- 以幂律（Power law）为信号，揭示几何图像的自相似性（self-similarity）

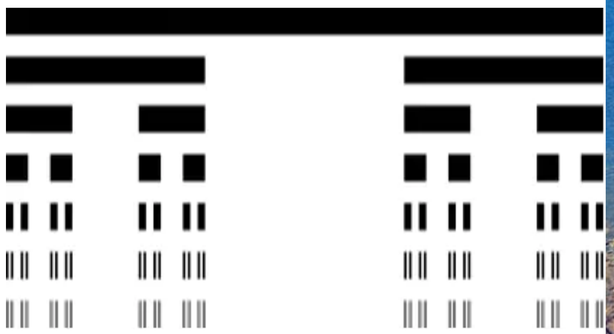
$$N(G) \sim G^{1-D}$$

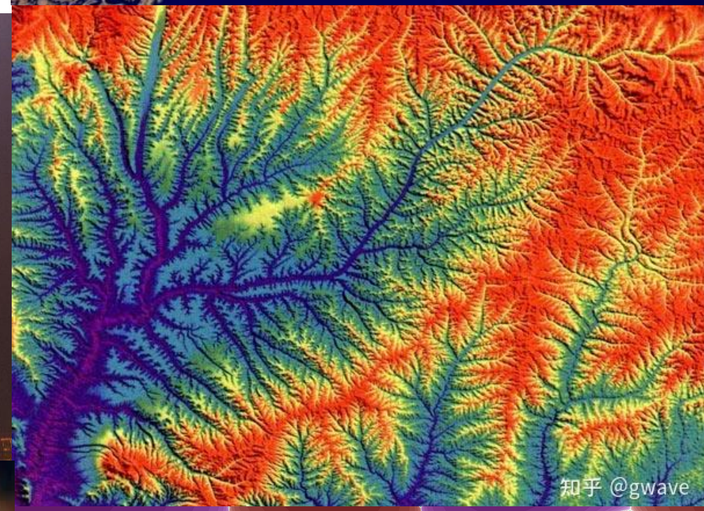


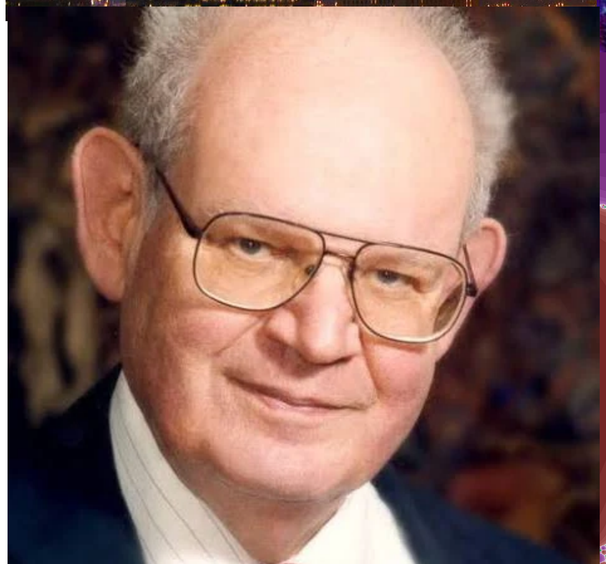
系统属性N

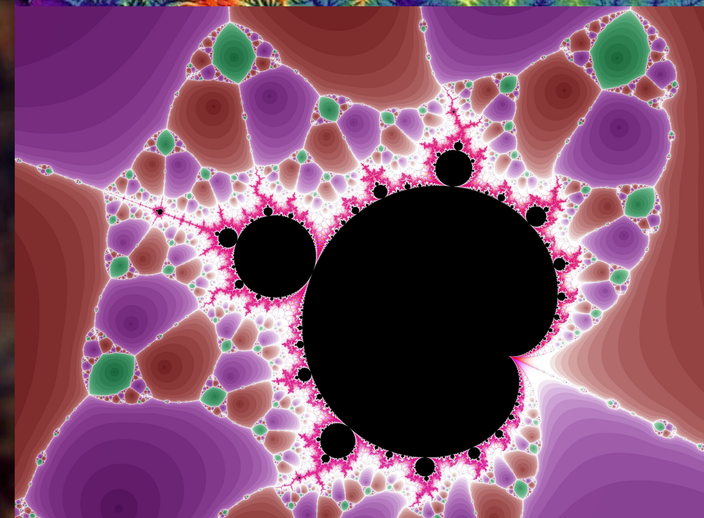
尺度G

Scaling

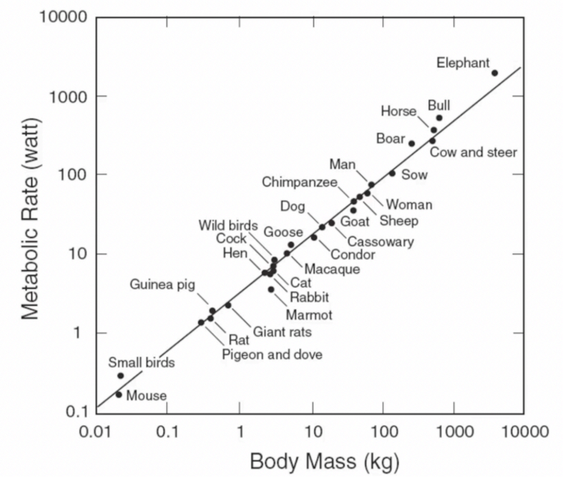




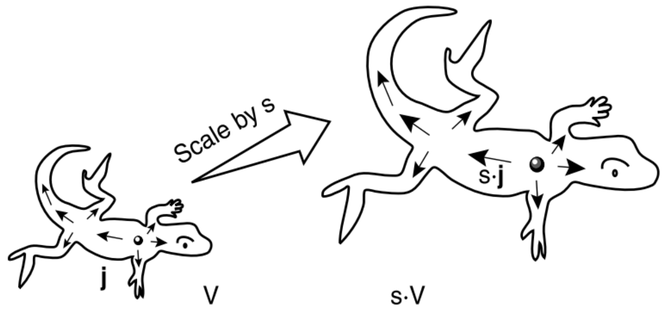


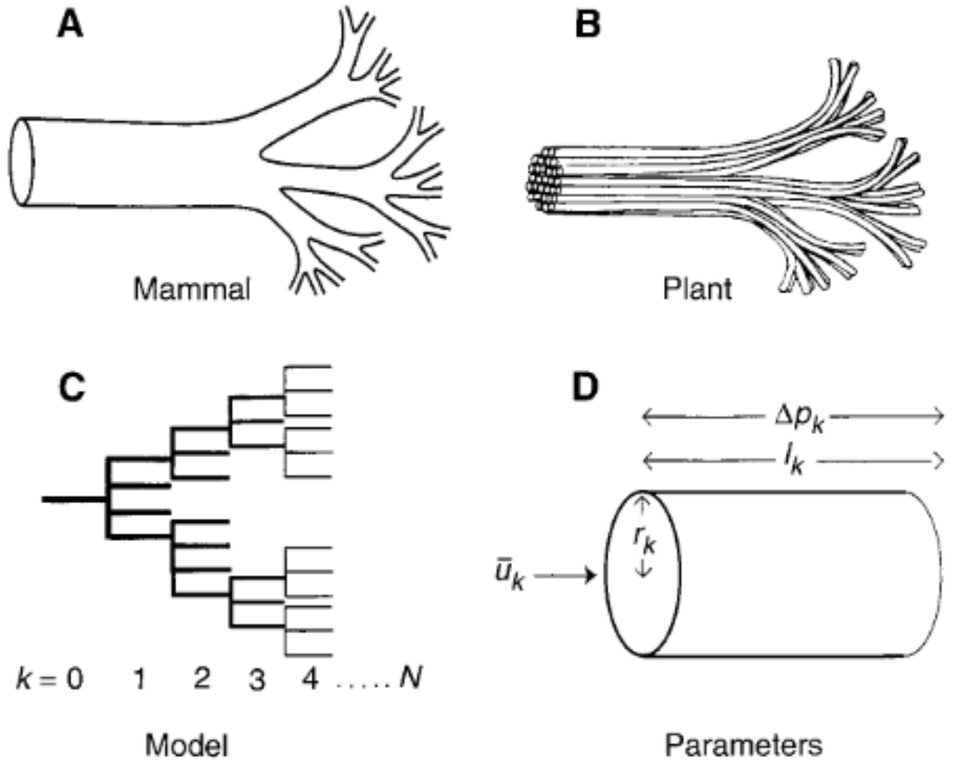
# 生物系统中的“分形”—克莱伯定律




$$F = cM^{3/4}$$

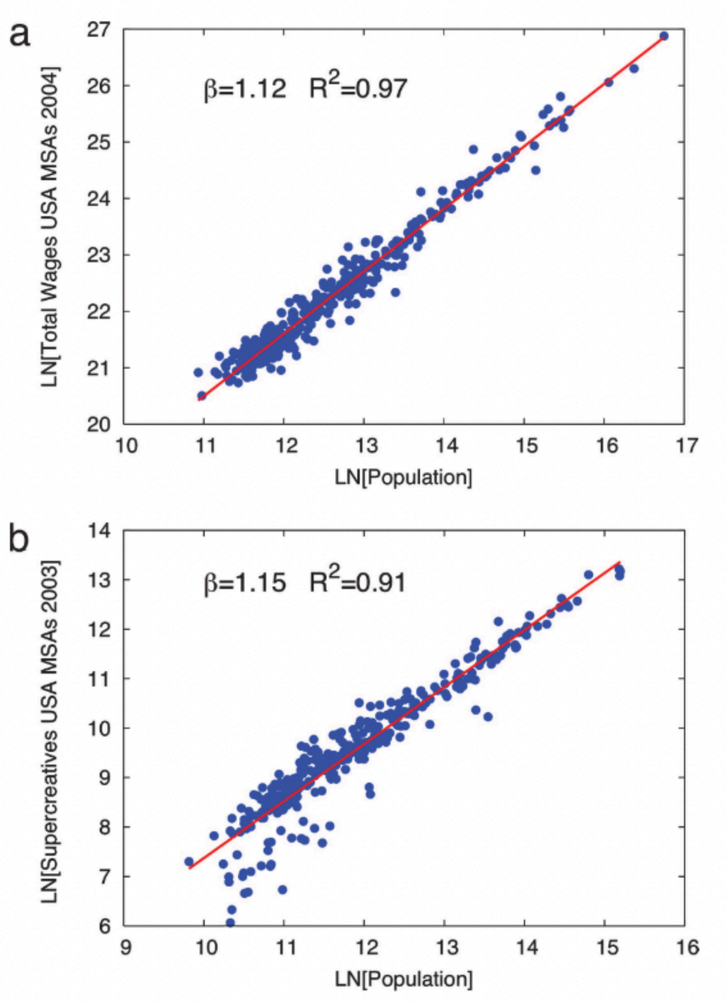
生物的代谢率和质量的关系

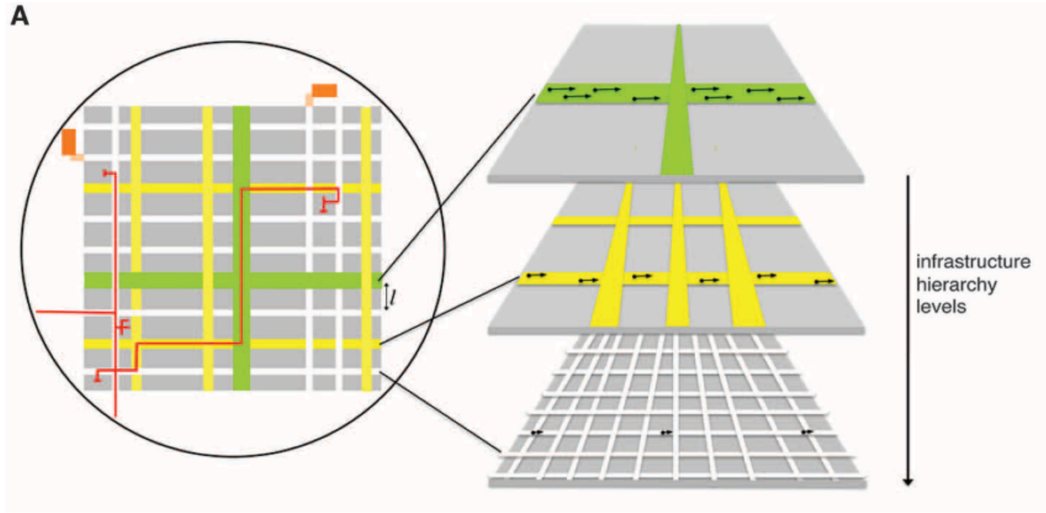


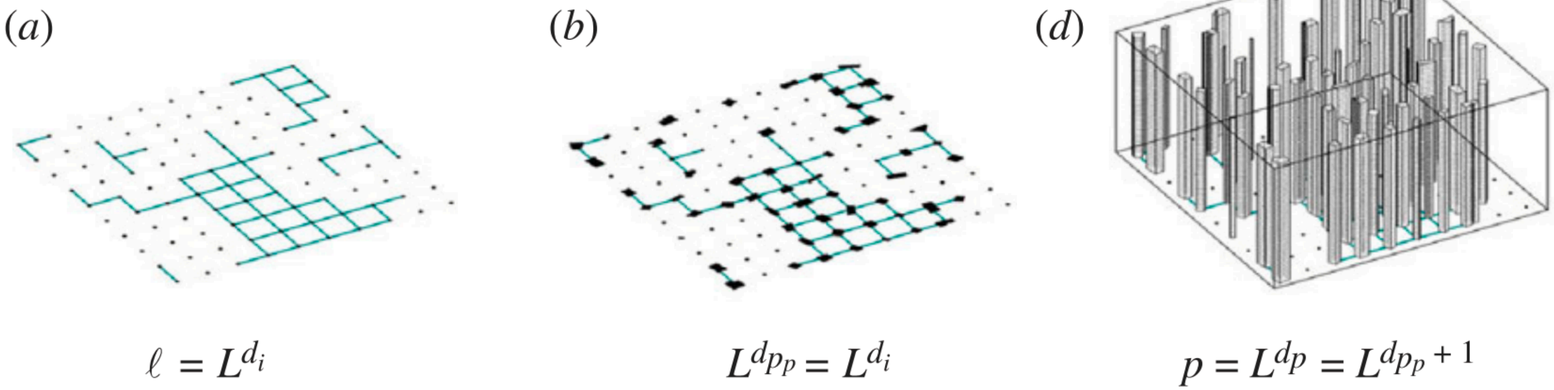




West, G. B., Brown, J. H., & Enquist, B. J. (1997). A general model for the origin of allometric scaling laws in biology. Science, 276(5309), 122–126.

# 城市中的“分形”



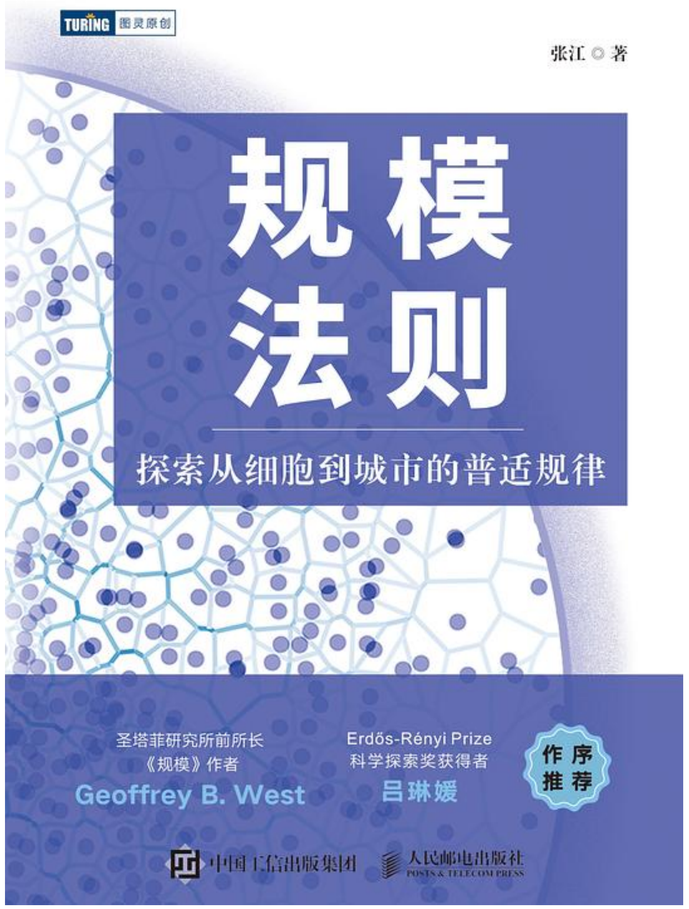


Bettencourt, L. M. A., Lobo, J., Helbing, D., Kühnert, C., & West, G. B. (2007). Growth, innovation, scaling, and the pace of life in cities. *PNAS*, *104*(17), 04

Molinero, C., & Thurner, S. (2021). How the geometry of cities determines urban scaling laws. Journal of the Royal Society Interface

Bettencourt, L. M. A. (2013). The origins of scaling in cities. Science, 340(6139), 1438–1441. https://doi.org/10.1126/science.1235823

# Scaling Theory




幂律、分形与自相似性

# Emergence

- 自下而上的建模
  - 复杂世界，简单规则

- 自上而下的计算
  - 从现象出发，研究涌现的量化

# Emergence

- **自下而上的建模**
  - **复杂世界，简单规则**
  - 优势：简洁，解释性强
  - 劣势：
    - 依赖科学家的经验
    - Case by Case，难以提炼出普适的建模方法

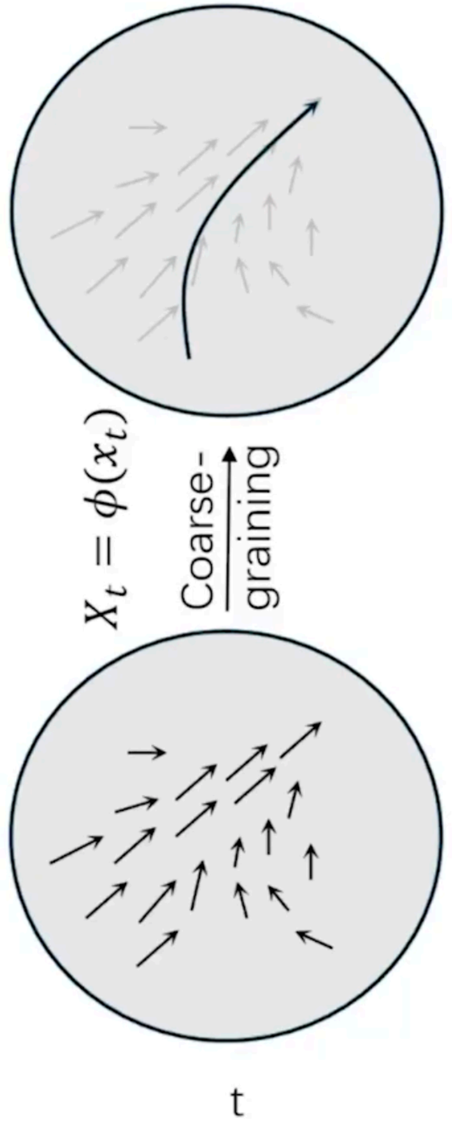
- **自上而下的计算**
  - **从现象出发，研究涌现的量化**
  - 优势：有希望实现理论统一性
  - 劣势：无法讨论微观机制

# Calculi of Emergence

鸟群的状态　　宏观X

粗粒化

单只鸟的状态　　微观x

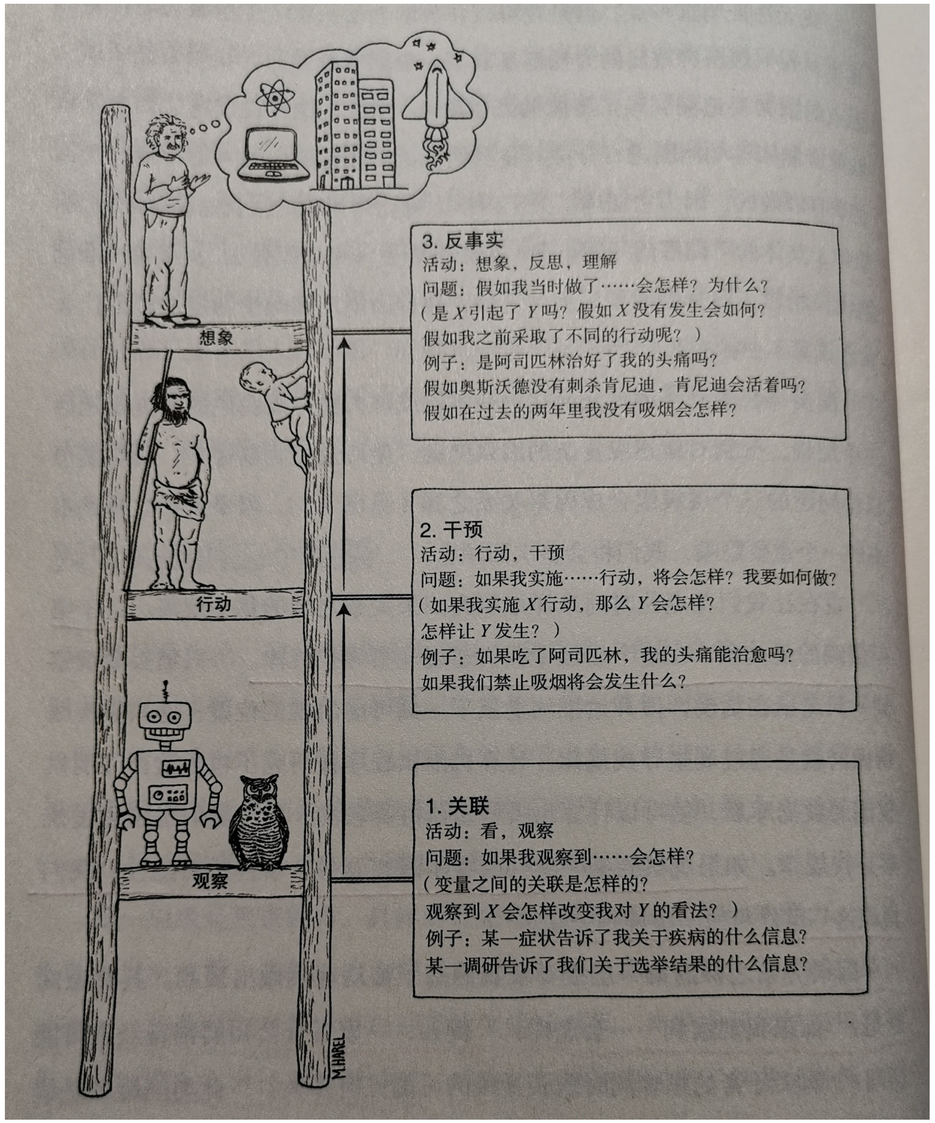


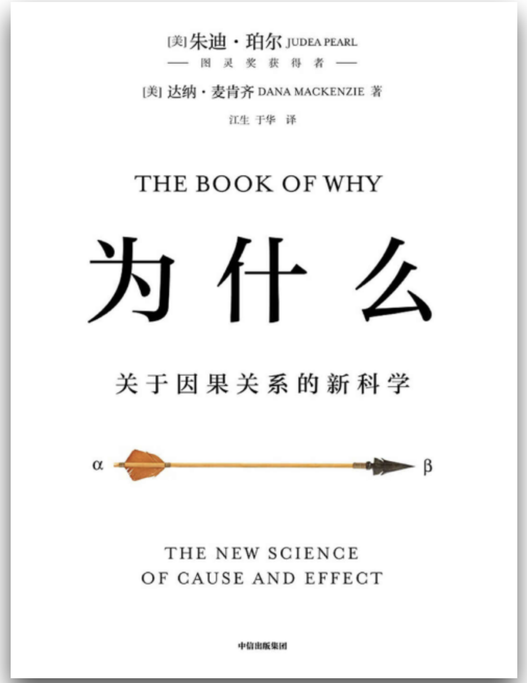

1. 如何粗粒化?

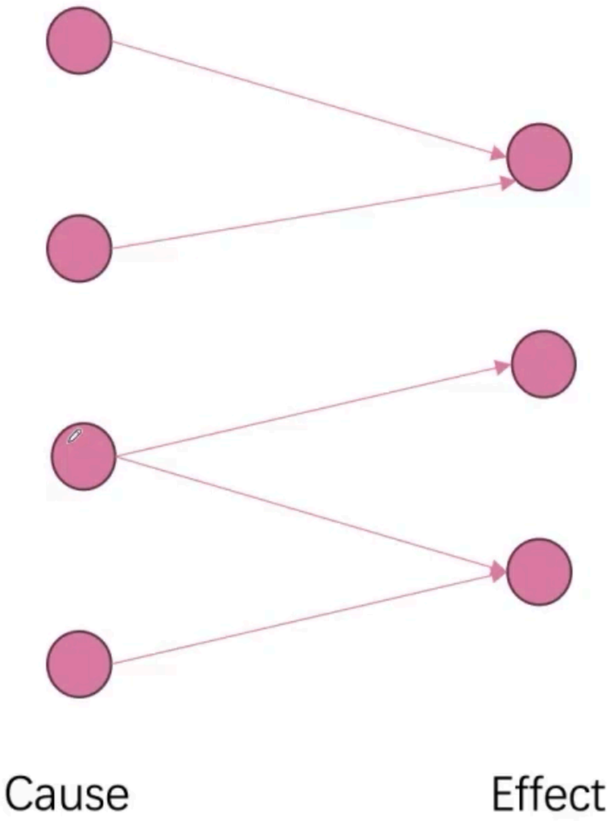
2. 宏观动力学和微观动力学的关系是什么?

因果性

# Emergence and Causality



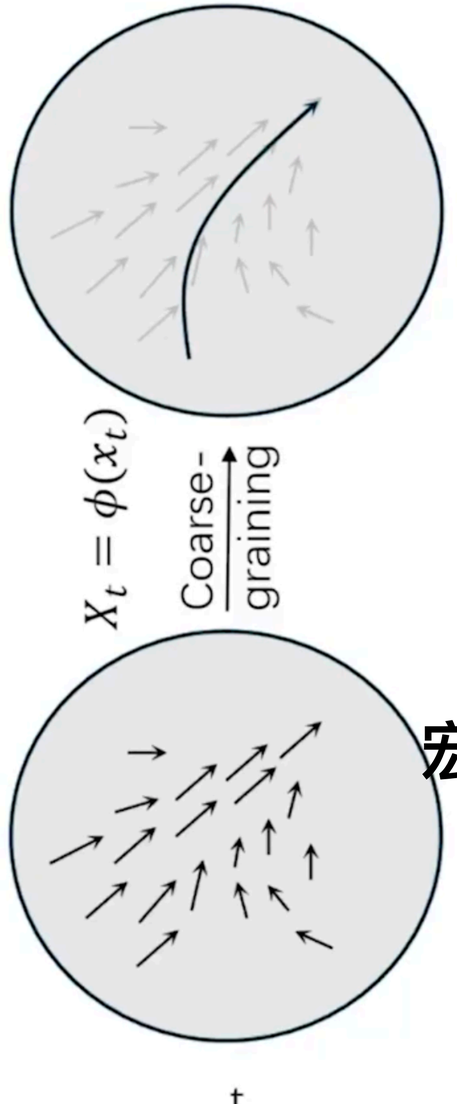



# Calculi of Emergence

鸟群的状态　　宏观X

粗粒化

单只鸟的状态　　微观x



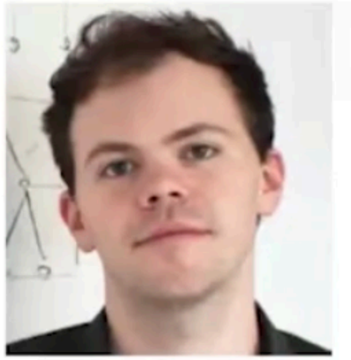

1. 如何粗粒化?

2. 宏观动力学和微观动力学的关系是什么?

因果性

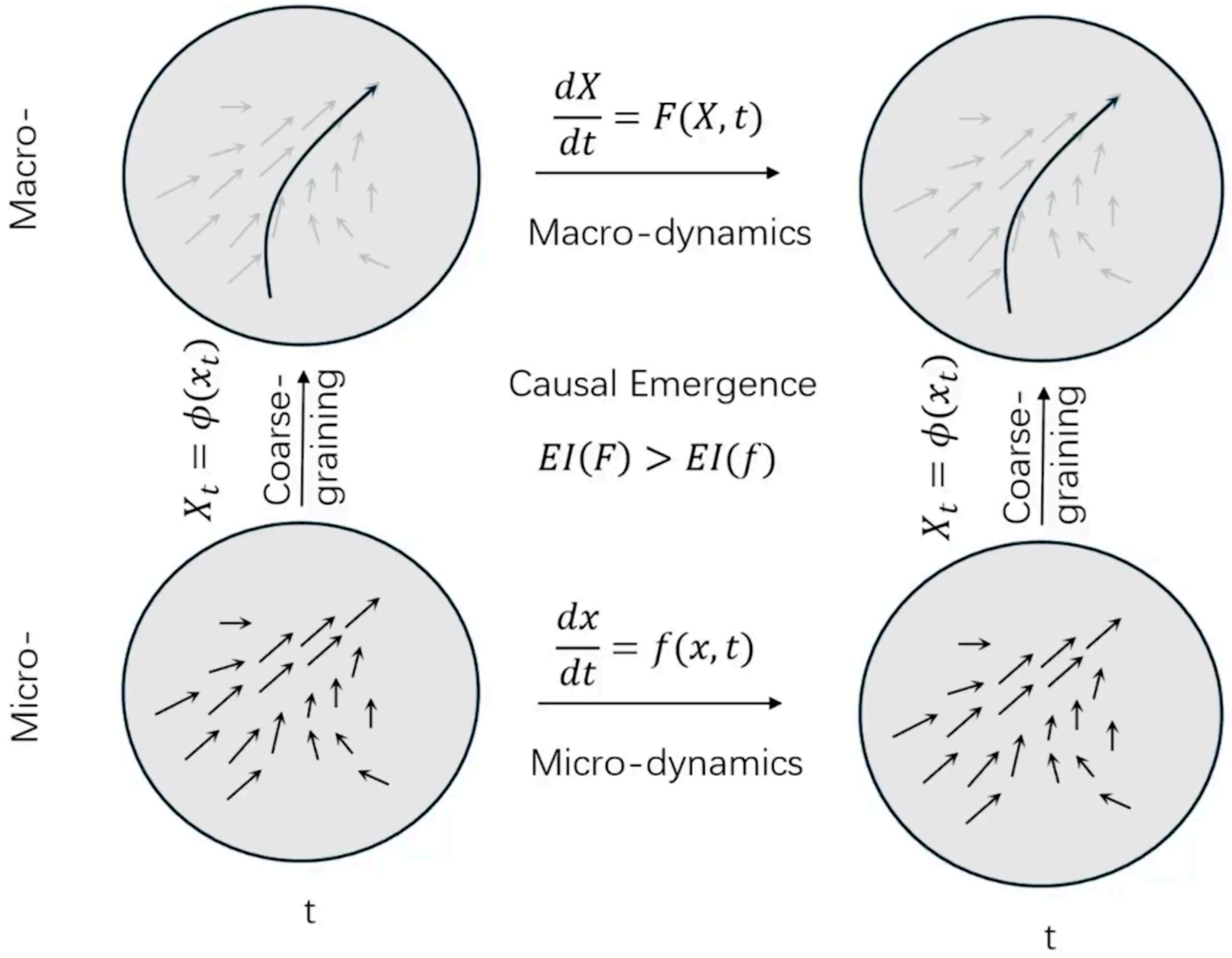
**宏观因果性 > 微观因果性**

# Causal Emergence Theory



Erik Hoel

Effective information （EI）



# Emergence

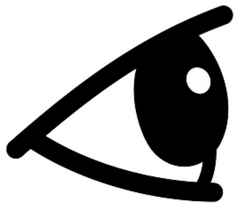
宏观因果性 > 微观因果性

因果性

宏观对称性 > 微观对称性

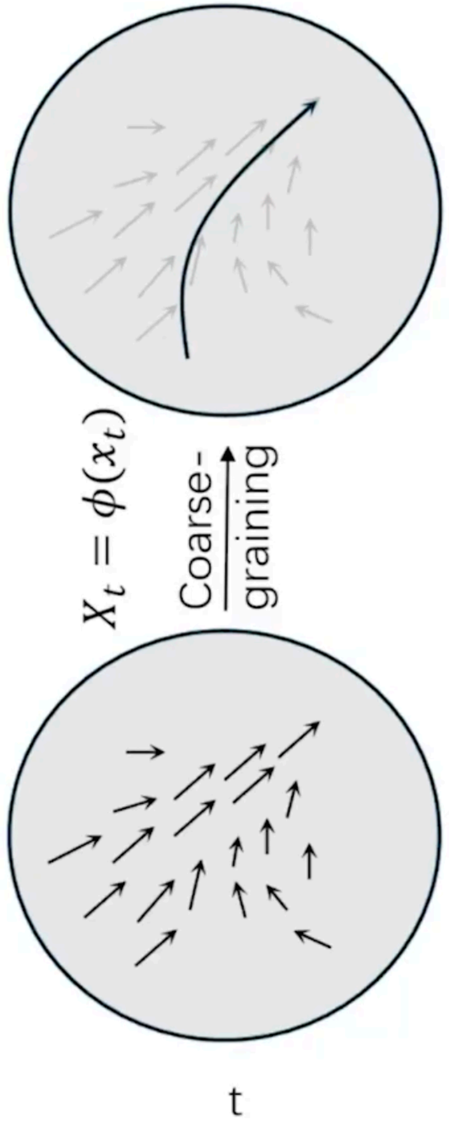
对称性

# Emergence



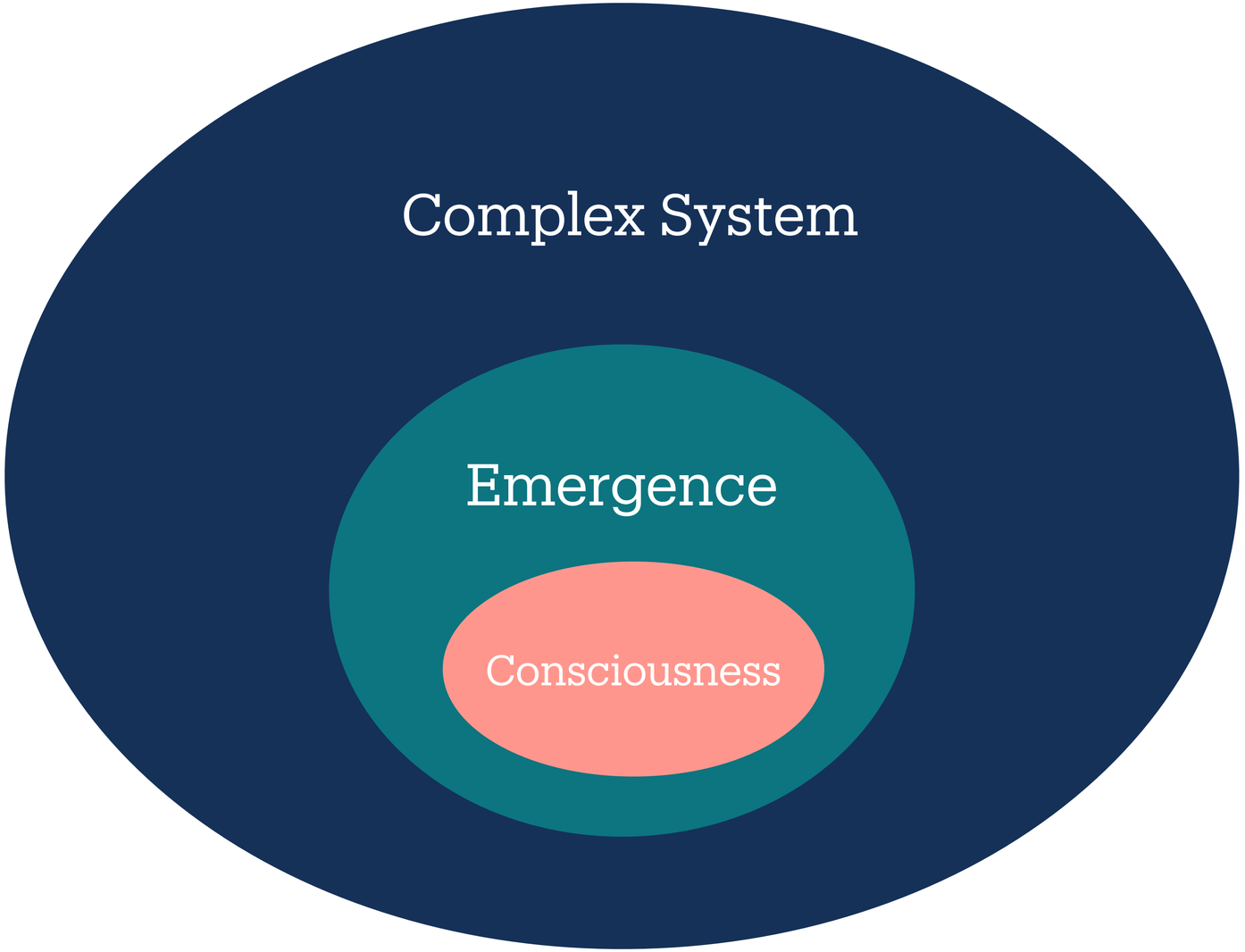
涌现不是在研究客观世界的性质

而是在研究观察者是如何理解世界



# Outline

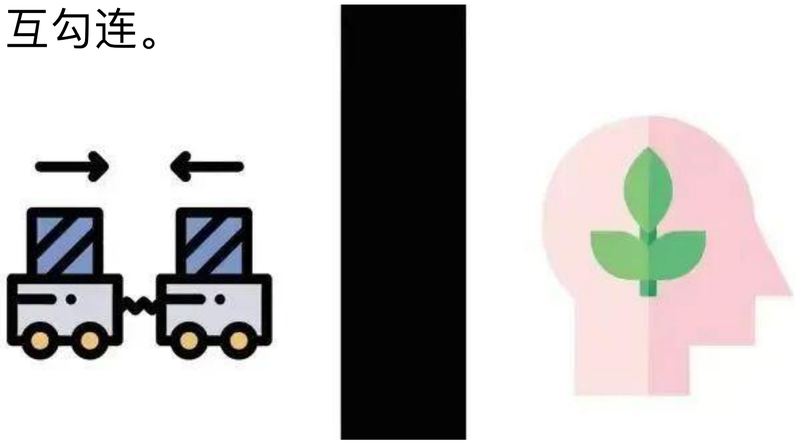
- **Complex Science**（15min）
- **Emergence**（45min）
  - Modeling from bottom to up
  - Measuring from up to bottom
- **Consciousness**（45min）
  - Evidences in neural science
  - Theory of Consciousness



# 意识问题

- 易问题（功能）：
  - 解释应用信息进行思考和行为的能力（被动）
  - 解释注意力、记忆、数据处理等问题（主动）
  - 寻找意识的神经相关物（neural correlates of consciousness, NCC）
- 难问题（体验）：
  - 为什么这些过程伴随着这种或那种特定的意识体验

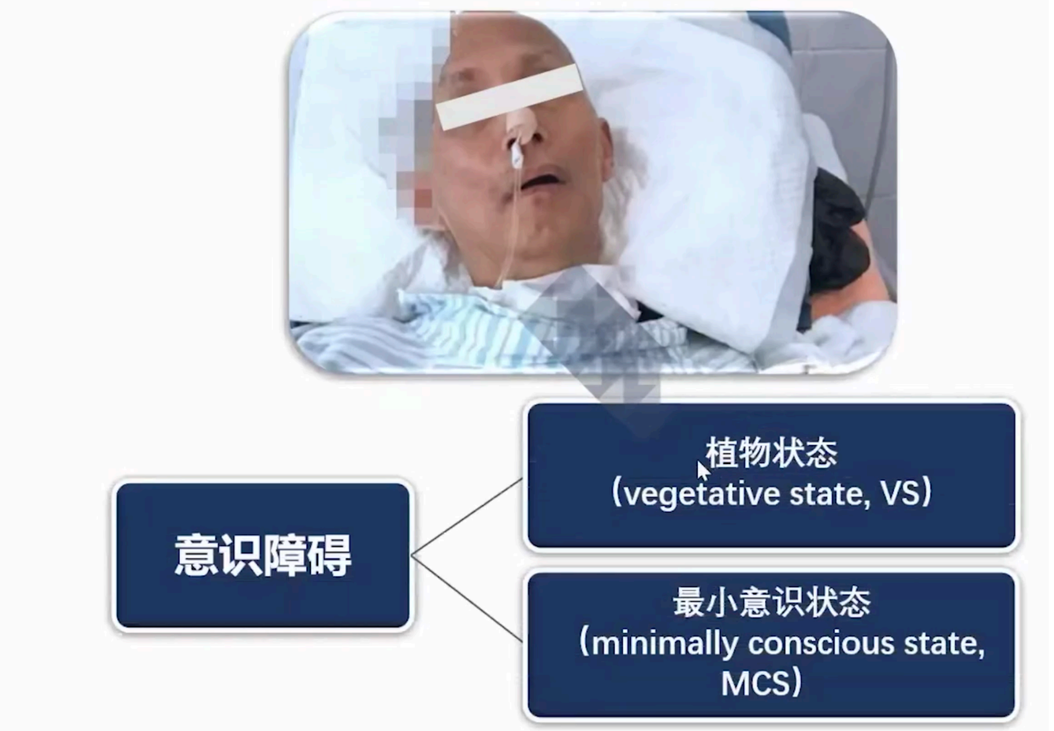
**莱布尼茨之堑：**观测到的现象，与知觉之间，似乎总有一个天堑一般，总是无法相互勾连。



The science of consciousness conference, 1994, Arizona

# 意识相关实证

- 睡眠
- 麻醉
- 儿童
- 死亡（濒死）
- 意识障碍诊断




**走向意识科学：从现象之窗到理论之梯**：2.意识的神经相关物研究：从临床到基础

狄海波

狄海波，杭州师范大学国际植物状态与意识科学研究所所长，国际意识障碍合作框架中方负责人，中华康复医学会意识障碍康复专业学会副主任委员，浙江省生理科学学会副理事长，比利时列日大学客座教授，浙江-比利时意识障碍联合实验室主任。杭州市第十届、十一届、十二届政协委员。国家自然科学基金重点项目终审专家。 目前主持国家自然科学基金重点国际（地区）合作研究项目1项，浙江省自然科学基金重点项目1项，军委后勤保障部重点项目子课题1项，曾主持意识障碍领域国家自然科学基金和 863 课题子项目4项、浙江省科技厅中外合作重大项目1项、欧盟基金1项等 十多项科研基金；发表意识障碍领域学术论文60 余篇，SCI 文章 30 篇，单篇 SCI引用450次。

# 如何测度意识

- 意识障碍诊断本质上是临床上的意识判定问题
  - 主观报告
  - 行为交流
  - 神经影像

**濒死体验内容(NDE-C)量表**

*我们希望你根据你在体验时（不是体验前，也不是体验后）的感受和想法来选择最合适的答案回答以下20项陈述（仅允许陈述一个答案）。*

*每种体验或感觉的强度不同，这就是为什么我们希望你使用下文对每项陈述所描述的评定等级(1-4)来确定你体验的强度。相反，如果你没有体验到陈述中所描述的现象，请选择"0-完全没有；无"。如果你在体验过程中多次体验到了相同现象，请考虑用最显著的现象进行回答。*

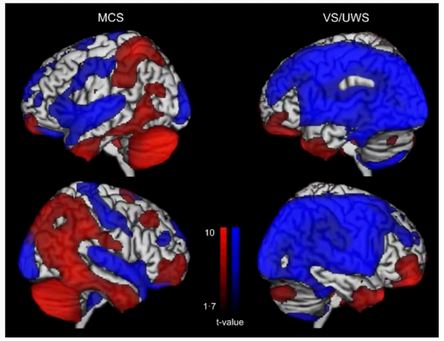
*评定等级:*
*0 - 完全没有；无*
*1 - 轻微*
*2 - 适中*
*3 -强烈；程度上等同于迄今生活中的任何其他强烈体验*
*4 - 极度强烈；超过我生命中任何其他时间并且比3强烈*

| | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| 1. 你对时间的感知改变了 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 2. 你的思考速度加快了 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 3. 你听到了没有任何物质化身的一个或几个声音 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 4. 你产生了对于自己、他人或宇宙顿悟的感觉 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 5. 你产生了平静和/或幸福的感觉 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 6. 你体会到了和谐或统一的感觉，仿佛你从属于一个更大的整体 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 7. 你看到或感觉到被一束没有任何确定物质来源的亮光包围着 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 8. 你体验到不同寻常的感觉（视觉、听觉、嗅觉、触觉和/或味觉） | ☐ | ☐ | ☐ | ☐ | ☐ |
| 9. 你觉知到了超出你的感觉通常能够感知到的事物 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 10. 你获得了对未来的深刻认识 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 11. 你有了处于自己身体之外，或与自己身体分离的印象 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 12 你有了离开尘世或进入新的维度和/或环境的感觉 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 13. 你看到或放下了你过去的事件 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 14. 你遇见了一个存在物和/或实体（它可能已消亡） | ☐ | ☐ | ☐ | ☐ | ☐ |
| 15. 你有了一种不存在、处于完全空虚和/或恐惧状态的感觉 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 16. 你走近到一个没有回头路的边界和/或地点 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 17. 你做出决定，或被迫，从这种体验中回来 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 18. 你有濒死和/或处于死亡状态的感觉 | ☐ | ☐ | ☐ | ☐ | ☐ |
| 19. 你看到或进入了一个通道（例如一条隧道或一扇门） | ☐ | ☐ | ☐ | ☐ | ☐ |
| 20. 你感到这种体验不能用语言充分描述 | ☐ | ☐ | ☐ | ☐ | ☐ |

**CRS-R量表**

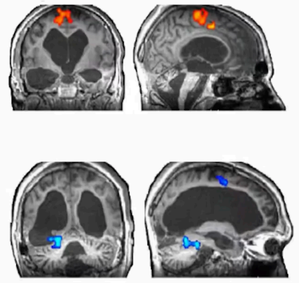
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | **听觉功能量表** | | | | | | | |
| + | 4 - 对指令有稳定反应 | | | | | | | |
| + | 3 - 可重复执行指令 | | | | | | | |
| | 2 - 声源定位 | | | | | | | |
| | 1 - 听觉惊吓反应 | | | | | | | |
| | 0 - 无 | | | | | | | |
| | **视觉功能量表** | | | | | | | |
| + | 5 - 识别物体 | | | | | | | |
| − | 4 - 物体定位：够向物体 | | | | | | | |
| − | 3 - 眼球追踪 | | | | | | | |
| − | 2 - 视觉对象定位 | | | | | | | |
| | 1 - 视觉惊吓反应 | | | | | | | |
| | 0 - 无 | | | | | | | |
| | **运动功能量表** | | | | | | | |
| * | 6 - 会使用物件 | | | | | | | |
| − | 5 - 自主性运动反应 | | | | | | | |
| − | 4 - 能摆弄物件 | | | | | | | |
| − | 3 - 对伤害性刺激定位 | | | | | | | |
| | 2 - 回撤屈曲 | | | | | | | |
| | 1 - 异常姿势 | | | | | | | |
| | 0 - 无/松弛 | | | | | | | |
| | **口部运动/言语功能量表** | | | | | | | |
| + | 3 - 言语表达可理解 | | | | | | | |
| | 2 - 发声/口部运动 | | | | | | | |
| | 1 - 反射性口部运动 | | | | | | | |
| | 0 - 无 | | | | | | | |
| | **交流评分量表** | | | | | | | |
| * | 2 - 交流完全准确 | | | | | | | |
| + | 1 - 交流不完全准确 | | | | | | | |
| | 0 - 无 | | | | | | | |
| | **觉醒水平评分量表** | | | | | | | |
| | 3 - 能注意 | | | | | | | |
| | 2 -无刺激下睁眼 | | | | | | | |
| | 1 - 刺激下睁眼 | | | | | | | |
| | 0 - 无睁眼 | | | | | | | |
| | **总分** | | | | | | | |
| | **诊断** | | | | | | | |

PET扫描局部代谢

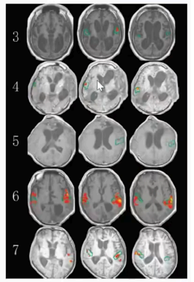



fMRI主动和被动范式

◆ 主动范式—想象打网球和空间移动



◆ 被动范式-唤名“鸡尾酒会效应”



# 意识理论

nature reviews neuroscience



Review Article | Published: 03 May 2022

## Theories of consciousness

Anil K. Seth ✉ & Tim Bayne

*Nature Reviews Neuroscience* **23**, 439–452 (2022) | Cite this article

**79k** Accesses | **539** Citations | **1250** Altmetric | Metrics

## Abstract

Recent years have seen a blossoming of theories about the biological and physical basis of consciousness. Good theories guide empirical research, allowing us to interpret data, develop new experimental techniques and expand our capacity to manipulate the phenomenon of interest. Indeed, it is only when couched in terms of a theory that empirical discoveries can ultimately deliver a satisfying understanding of a phenomenon. However, in

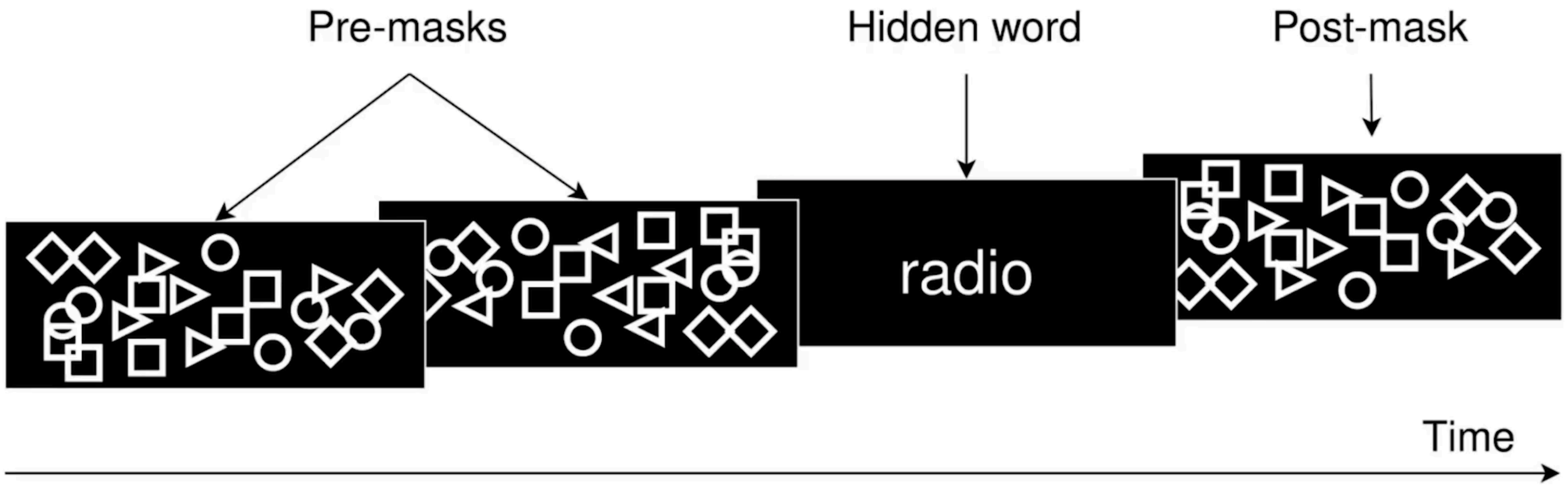
全局工作空间理论 vs 整合信息论

Table 1 | **A selection of theories of consciousness**

| Theory | Primary claim | Key refs |
|---|---|---|
| Higher-order theory (HOT) | Consciousness depends on meta-representations of lower-order mental states | 31,46 |
| Self-organizing meta-representational theory | Consciousness is the brain's (meta-representational) theory about itself | 34,140 |
| Attended intermediate representation theory | Consciousness depends on the attentional amplification of intermediate-level representations | 141,142 |
| Global workspace theories (GWTs) | Consciousness depends on ignition and broadcast within a neuronal global workspace where fronto-parietal cortical regions play a central, hub-like role | 47–49 |
| Integrated information theory (IIT) | Consciousness is identical to the cause–effect structure of a physical substrate that specifies a maximum of irreducible integrated information | 57,59,60 |
| Information closure theory | Consciousness depends on non-trivial information closure with respect to an environment at particular coarse-grained scales | 143 |
| Dynamic core theory | Consciousness depends on a functional cluster of neural activity combining high levels of dynamical integration and differentiation | 144 |
| Neural Darwinism | Consciousness depends on re-entrant interactions reflecting a history of value-dependent learning events shaped by selectionist principles | 145,146 |
| Local recurrency | Consciousness depends on local recurrent or re-entrant cortical processing and promotes learning | 65,71 |
| Predictive processing | Perception depends on predictive inference of the causes of sensory signals; provides a framework for systematically mapping neural mechanisms to aspects of consciousness | 67,73,79 |
| Neuro-representationalism | Consciousness depends on multilevel neurally encoded predictive representations | 84 |
| Active inference | Although views vary, in one version consciousness depends on temporally and counterfactually deep inference about self-generated actions | 76; see also 91 |
| Beast machine theory | Consciousness is grounded in allostatic control-oriented predictive inference | 13,75,77; see also 90 |
| Neural subjective frame | Consciousness depends on neural maps of the bodily state providing a first-person perspective | 24 |
| Self comes to mind theory | Consciousness depends on interactions between homeostatic routines and multilevel interoceptive maps, with affect and feeling at the core | 23,147 |
| Attention schema theory | Consciousness depends on a neurally encoded model of the control of attention | 148 |
| Multiple drafts model | Consciousness depends on multiple (potentially inconsistent) representations rather than a single, unified representation that is available to a central system | 149 |
| Sensorimotor theory | Consciousness depends on mastery of the laws governing sensorimotor contingencies | 88 |
| Unlimited associative learning | Consciousness depends on a form of learning which enables an organism to link motivational value with stimuli or actions that are novel, compound and non-reflex inducing | 150 |
| Dendritic integration theory | Consciousness depends on integration of top-down and bottom-up signalling at a cellular level | 151 |
| Electromagnetic field theory | Consciousness is identical to physically integrated, and causally active, information encoded in the brain's global electromagnetic field | 152 |
| Orchestrated objective reduction | Consciousness depends on quantum computations within microtubules inside neurons | 18 |

Our selection of theories includes those that are either neurobiological in nature or potentially expressible in neurobiological terms.

# 全局工作空间理论

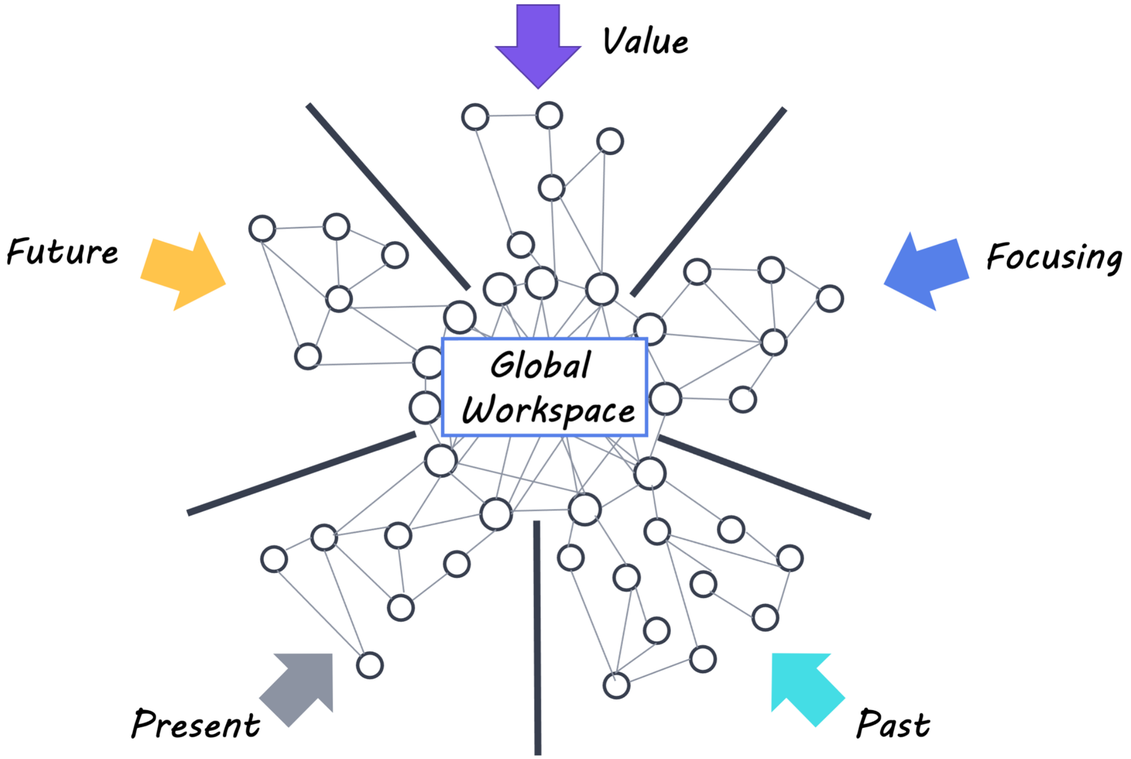



意识的进入存在一个阈值

无意识也在进行信息处理

Dehaene, S., & Changeux, J. (2011). Review Experimental and Theoretical Approaches to Conscious Processing. Neuron, 70(2), 200–227. https://doi.org/10.1016/j.neuron.2011.03.018

# 全局工作空间理论




Stanislas Dehaene

Bernard Baars

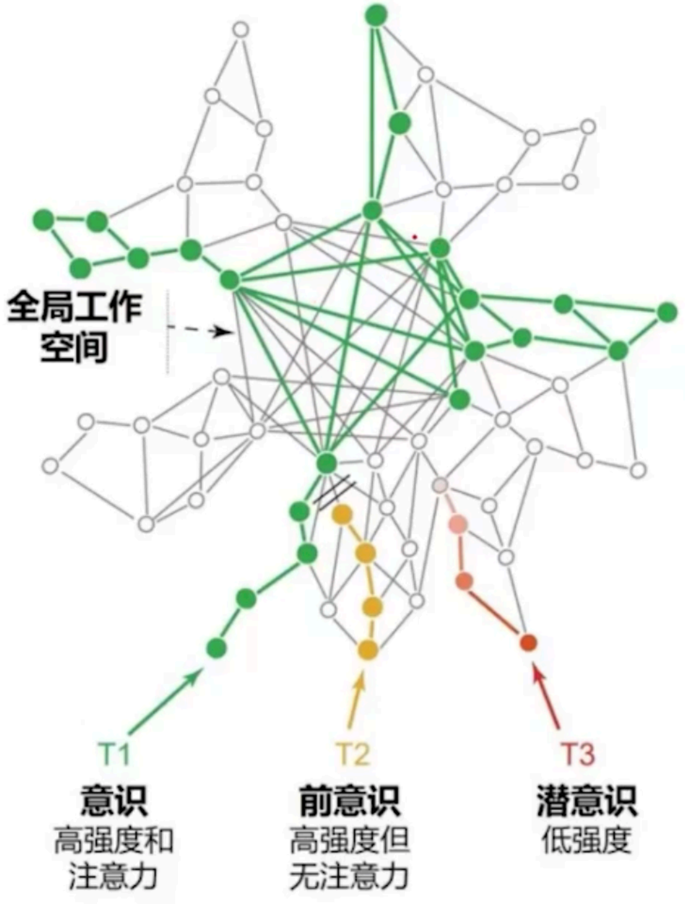
Jean-Pierre Changeux

当感觉输入、记忆等受到关注时，他们的信息就会进入全局工作空间，供各个认知过程访问。

比如开车：周围的建筑物，红绿灯

还是从功能的角度回答什么情况下是“有意识”

# 全局工作空间理论




一个不严谨的类比：小世界网络

刺激较小时，不激活

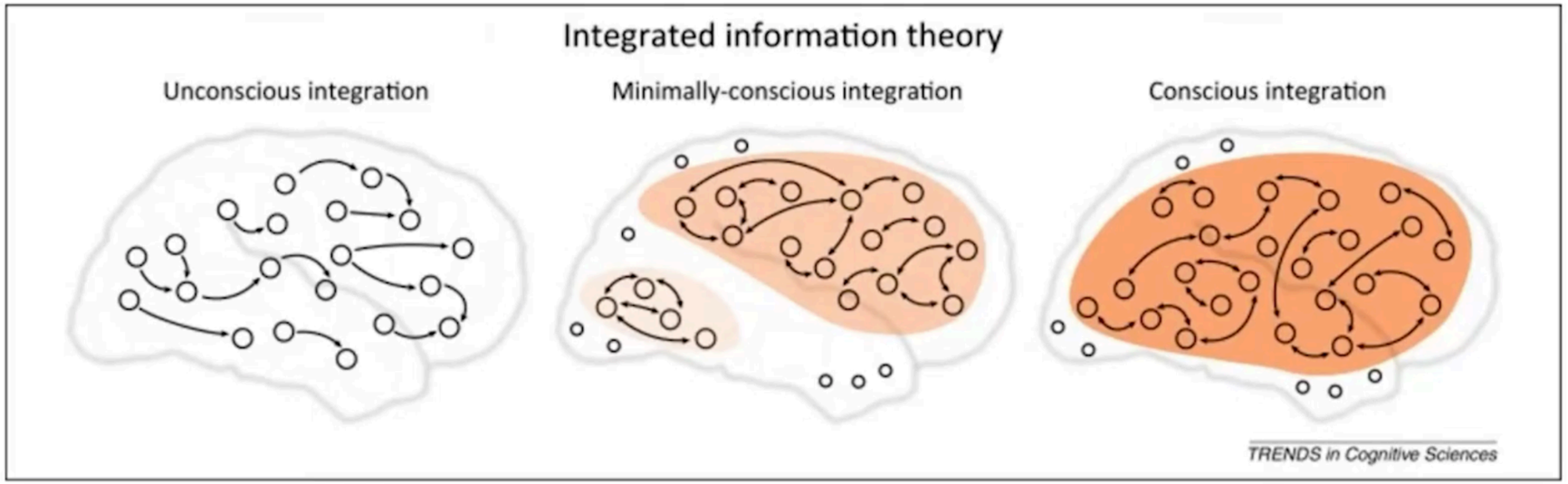
增大刺激后会出现全局激活的临界现象

全局工作空间主要在前额叶部分

观测——>理论建模——>实证验证

Dehaene, S., & Changeux, J. (2011). Review Experimental and Theoretical Approaches to Conscious Processing. Neuron, 70(2), 200–227. https://doi.org/10.1016/j.neuron.2011.03.018

# 整合信息论



Giulio Tononi

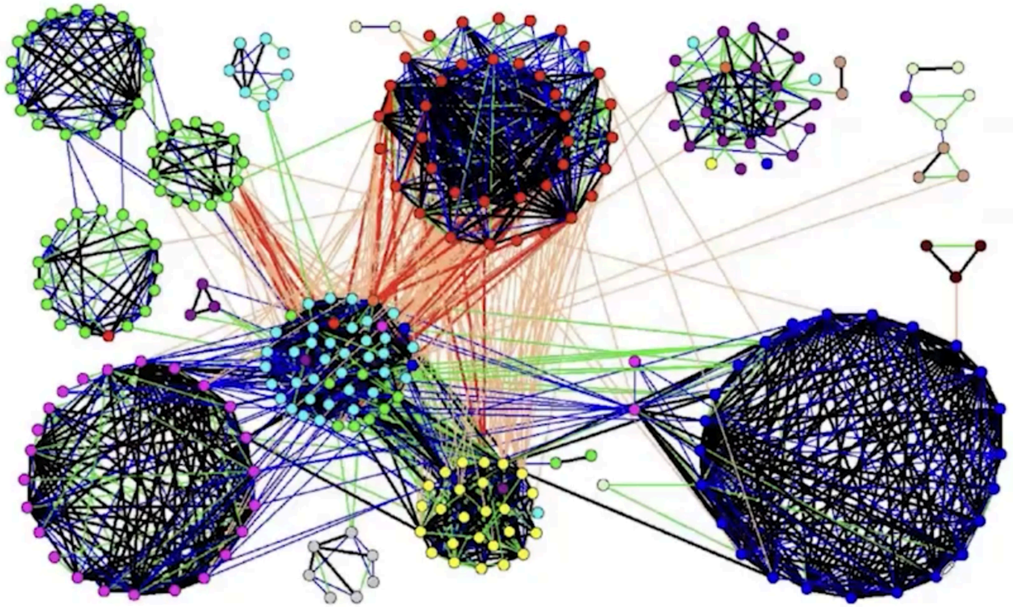
Pre IIT → IIT1.0 → IIT2.0 → IIT3.0 → IIT4.0

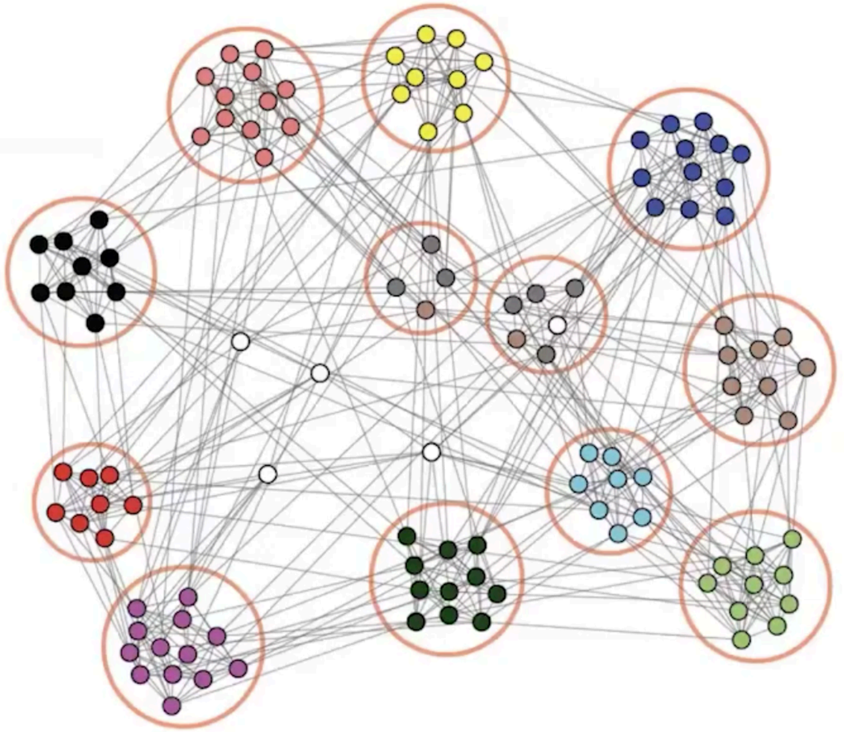
TONONI 94 TONONI 03 TONONI 08 TONONI 14 TONONI 23

# 整合信息论

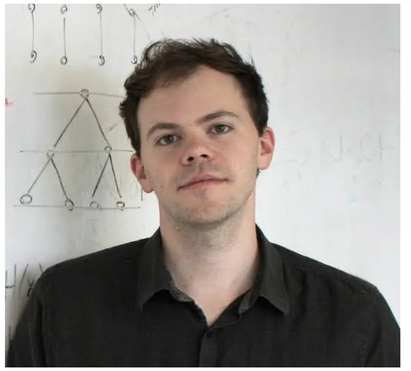
$\Phi$

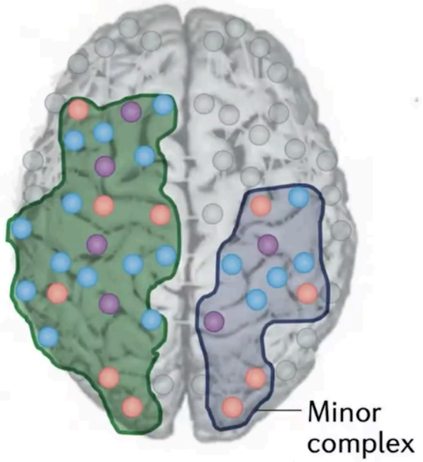
The degree of information integration for distributed systems

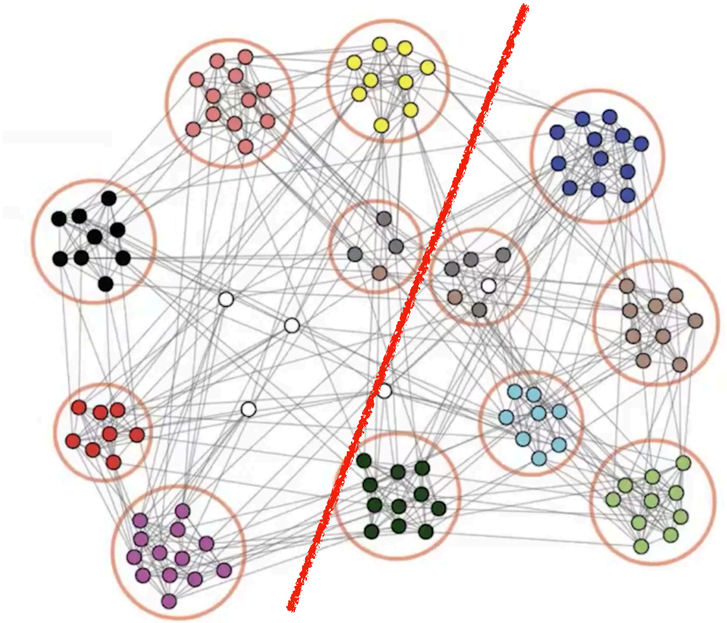




# 整合信息论



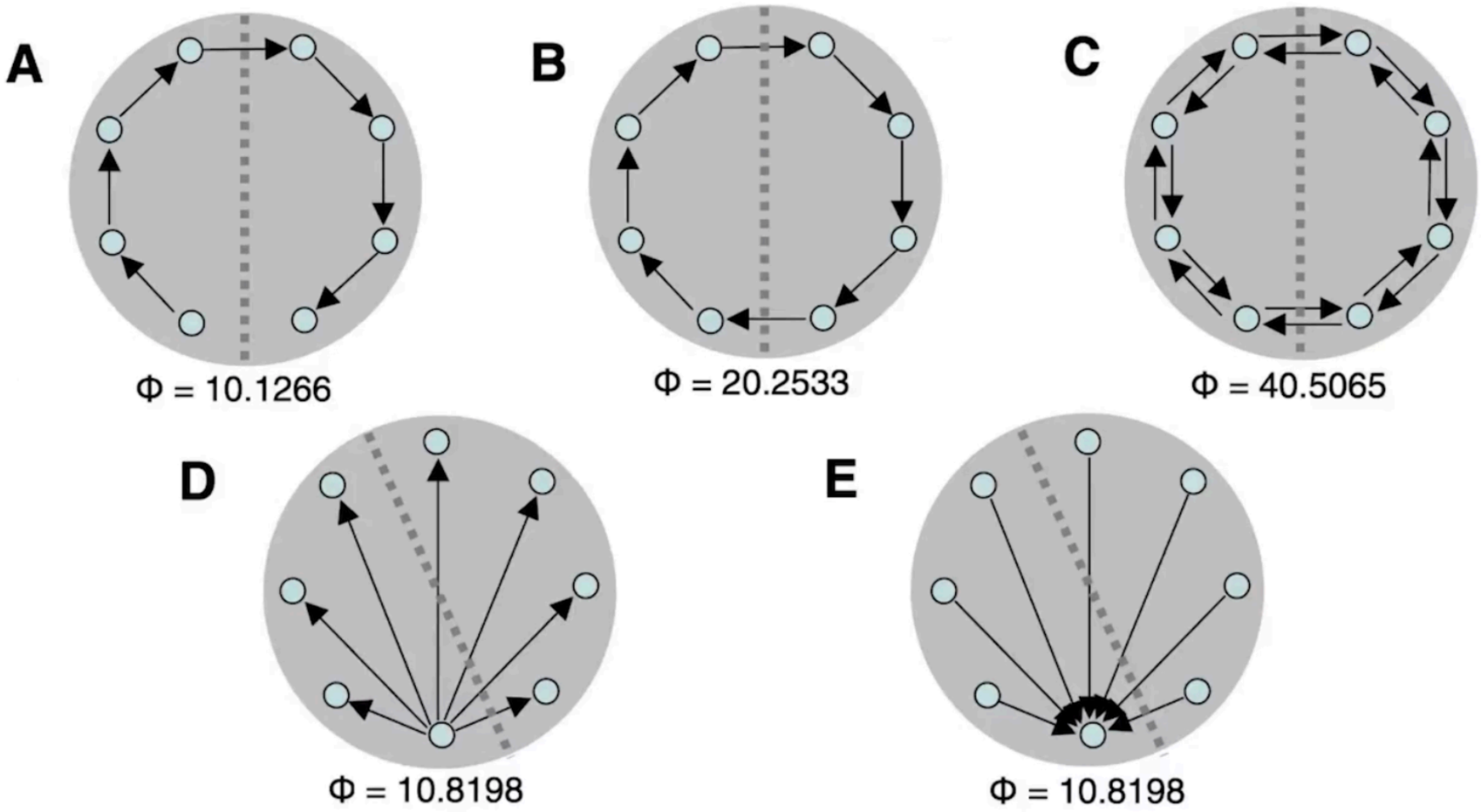


Φ的计算

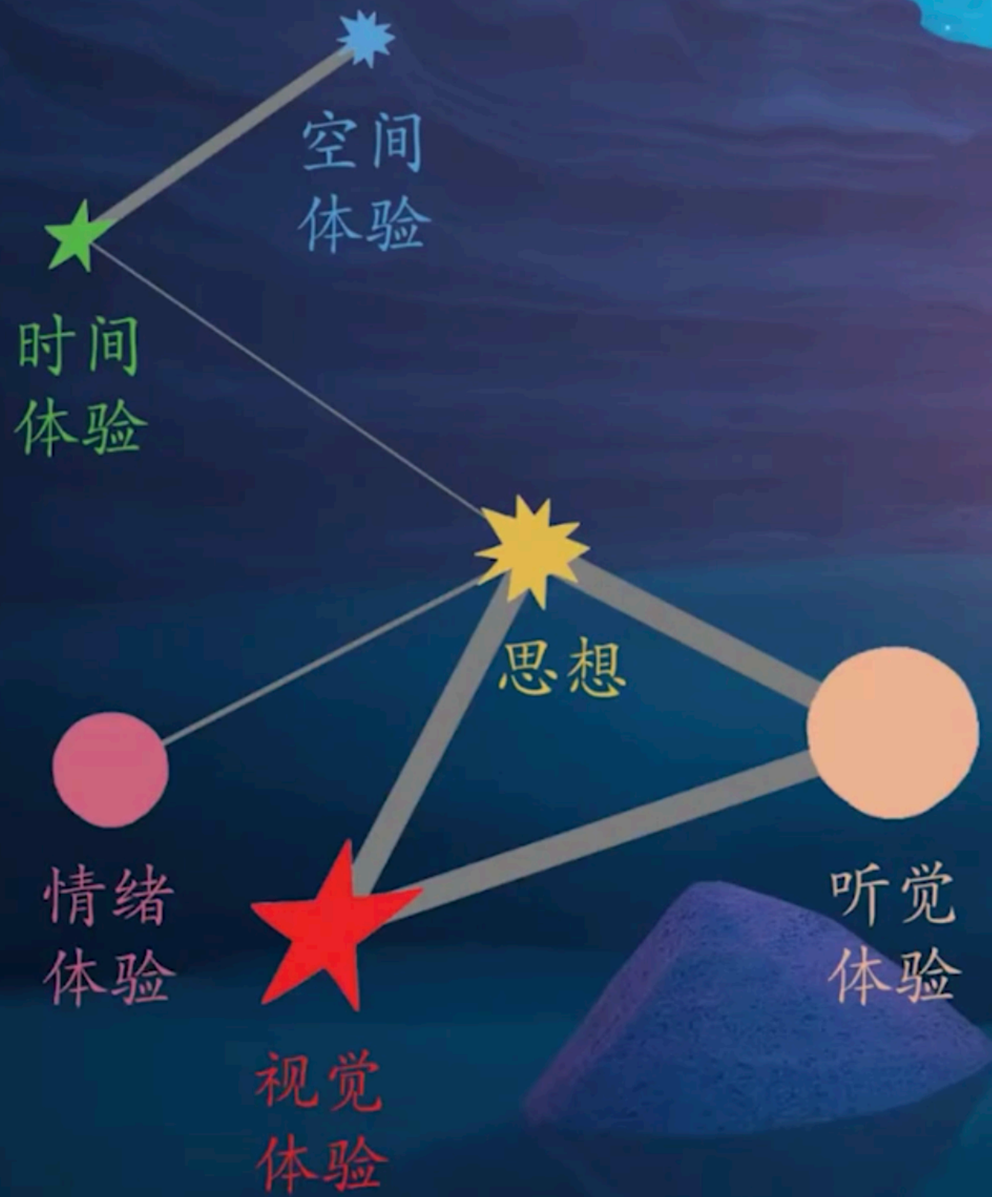
1. 对系统划分子集
2. 对每一个子集做“切割”，每次切割能得到一个Effective Information值
3. 遍历切割方式，找到最小的EI值作为当前子集的$\Phi_s$
4. 取所有子集中的最大值作为最终的Φ，即$\Phi(X) = max_{s \in X}\{\Phi_s\}$

# 整合信息论




Tononi G, Sporns O. Measuring information integration. BMC Neurosci. 2003

# 何为意识？



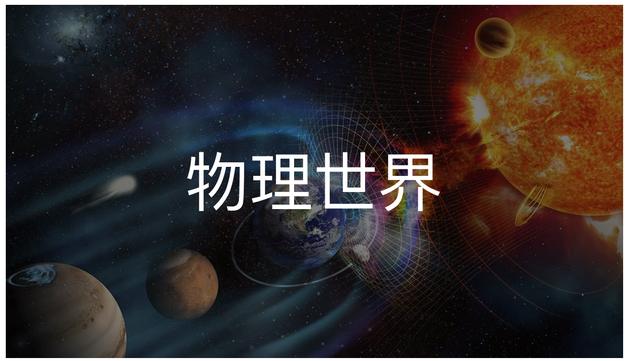

想象你在海边观看日落
你所体验到的
太阳的温暖、天空的色彩
沙子的细腻、海风的吹拂
以及你的情绪、思想等等
这些都属于你的意识
共同构成了那个时刻
独一无二的体验

意识即体验

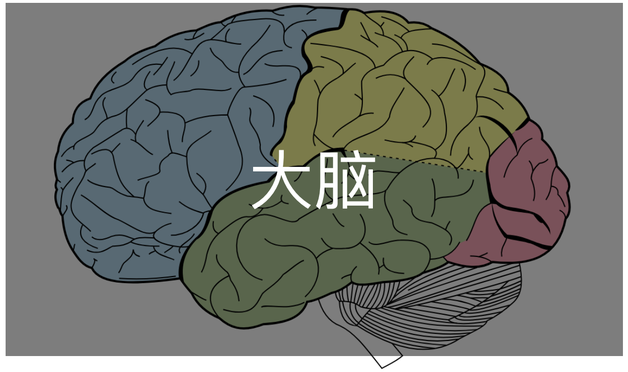
Consciousness is
anything and everything
one experiences.
(Koch, 2018, Nature)

# 整合信息论背后的价值观

外在视角




独立于意识而存在

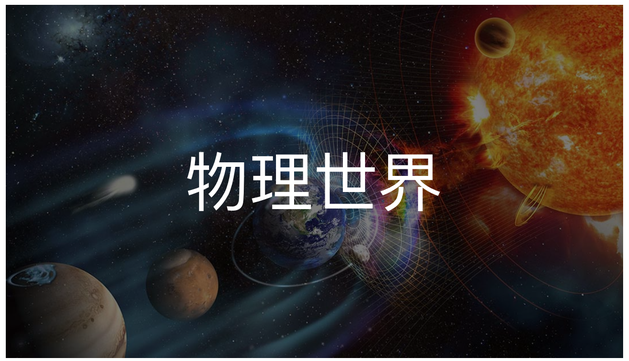



由功能定义其存在

是大脑的一种功能性，而不是一个实体存在

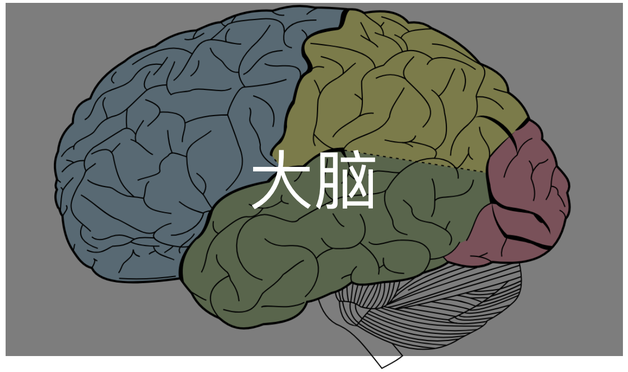
# 整合信息论背后的价值观

内在视角（体验者视角）




无法被完全客观观察，只能被意识主观体验

因为“观察”本身就是一种意识体验




由内在结构决定

而不是由其执行的功能决定

是有自身结构的实体性存在

是物理系统“内在结构”的体验化呈现

# 整合信息论公理体系

第零公理：**Existence**，体验是存在的

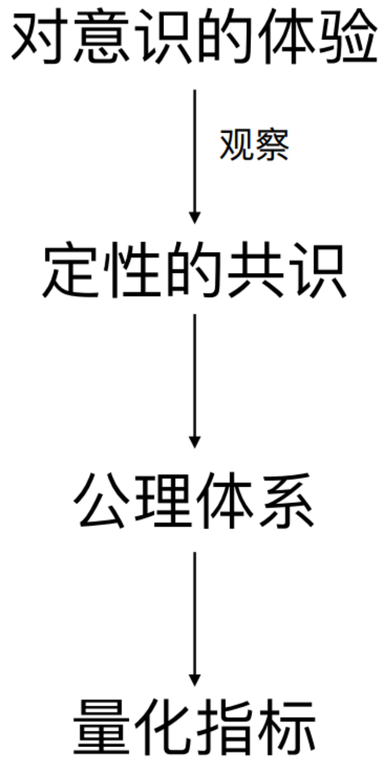
第一公理：**Intrinsicality**，体验是内在的，它只为你一个人存在

第二公理：**Information**，体验是特定的

第三公理：**Integration**，体验是整合的

第四公理：**Exclusion**，体验是确定的

第五公理：**Composition**，体验是有结构的



# 整合信息论

存在性 = 因果力

内在结构 = 因果结构

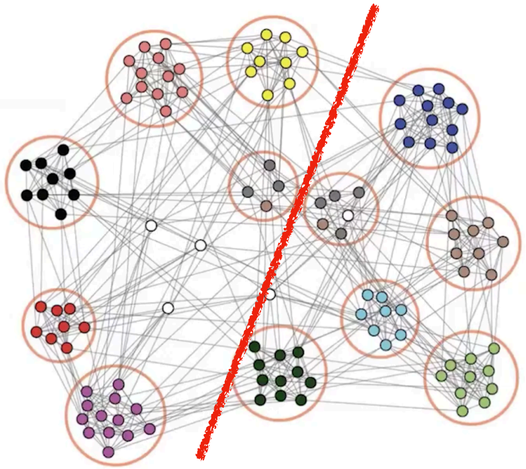
对意识的体验

观察

定性的共识

公理体系

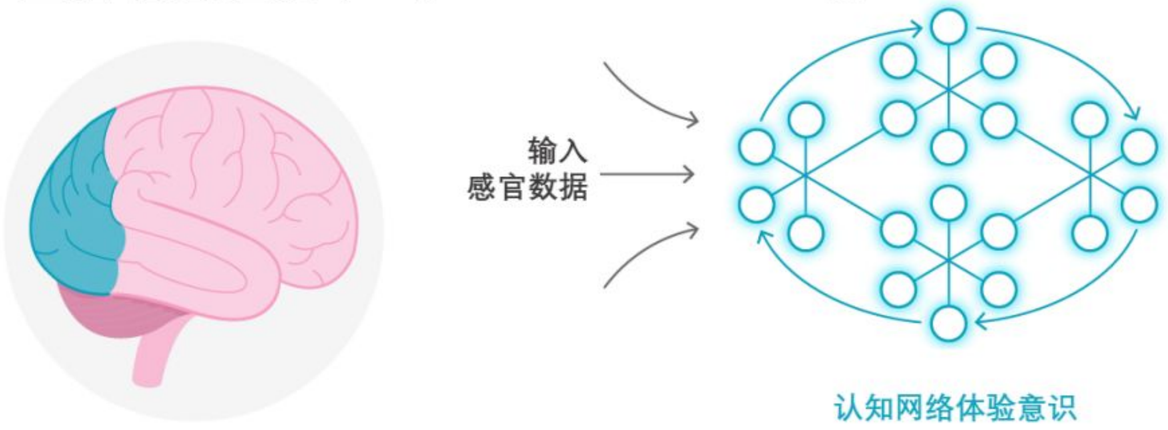
量化指标



| | | |
|---|---|---|
| 结构不同 | 体验不同 | phi不同 |
| 状态不同 | 体验不同 | phi不同 |
| 功能相同 | 体验不同 | phi不同 |

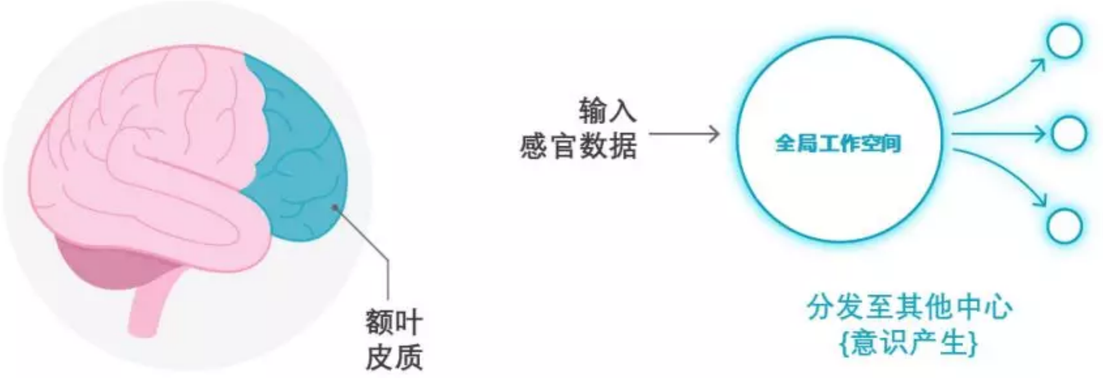
# GNWT vs IIT

## 整合信息理论 (Integrated Information Theory)




整合信息理论认为，意识是认知网络固有的、对自身施加“因果能力”。大脑的后部可能存有适合这种能力的结构。

## 全局空间理论（Global Workspace Theory）

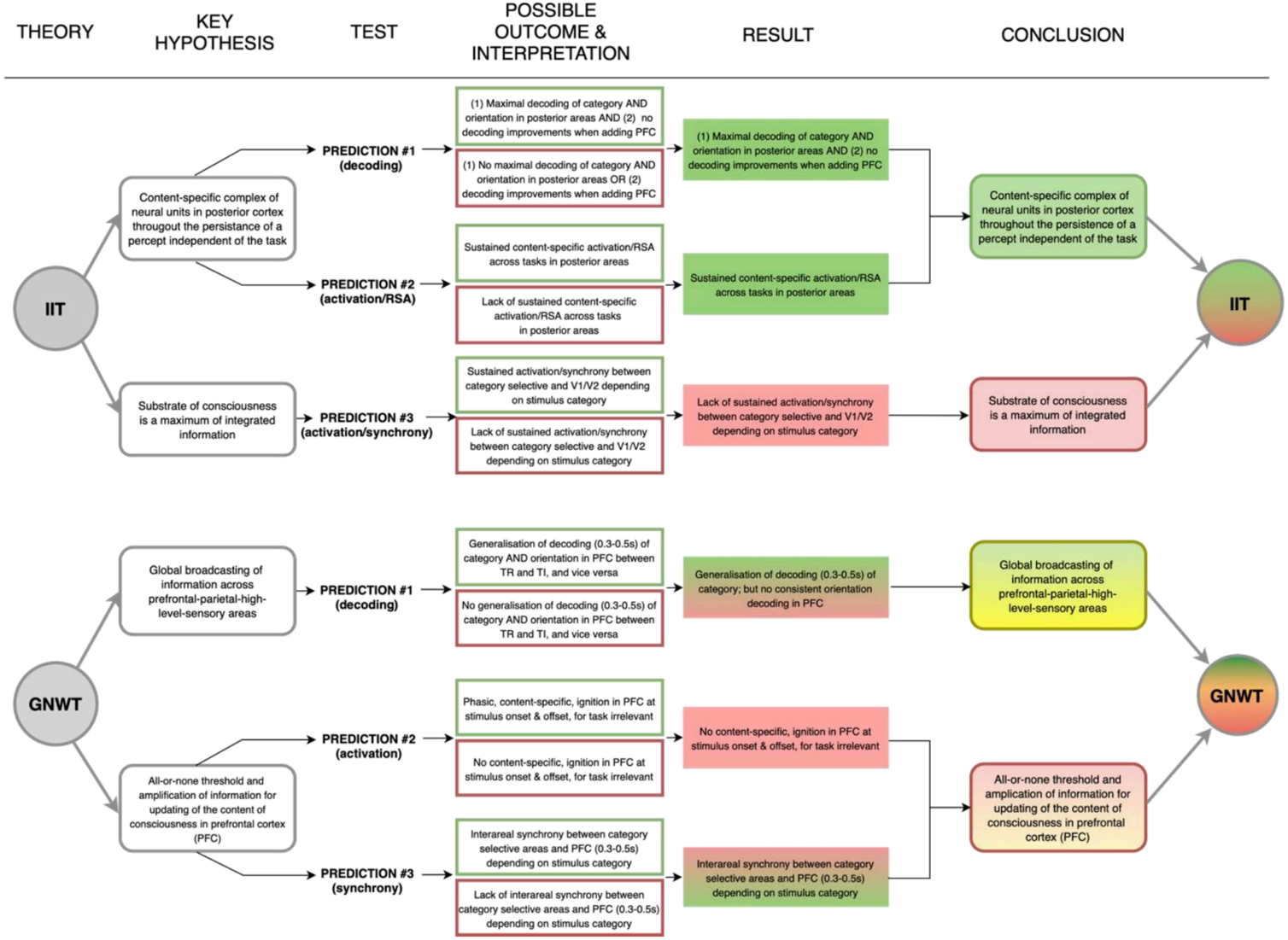



这一理论认为，意识是信息处理的一种形式，当某种体验的感官数据进入“全局工作空间”并分发到其他中心时，就会产生“意识”。产生这个过程的结构可能在大脑额叶皮层。

# GNWT vs IIT

**An adversarial collaboration to critically evaluate theories of consciousness**

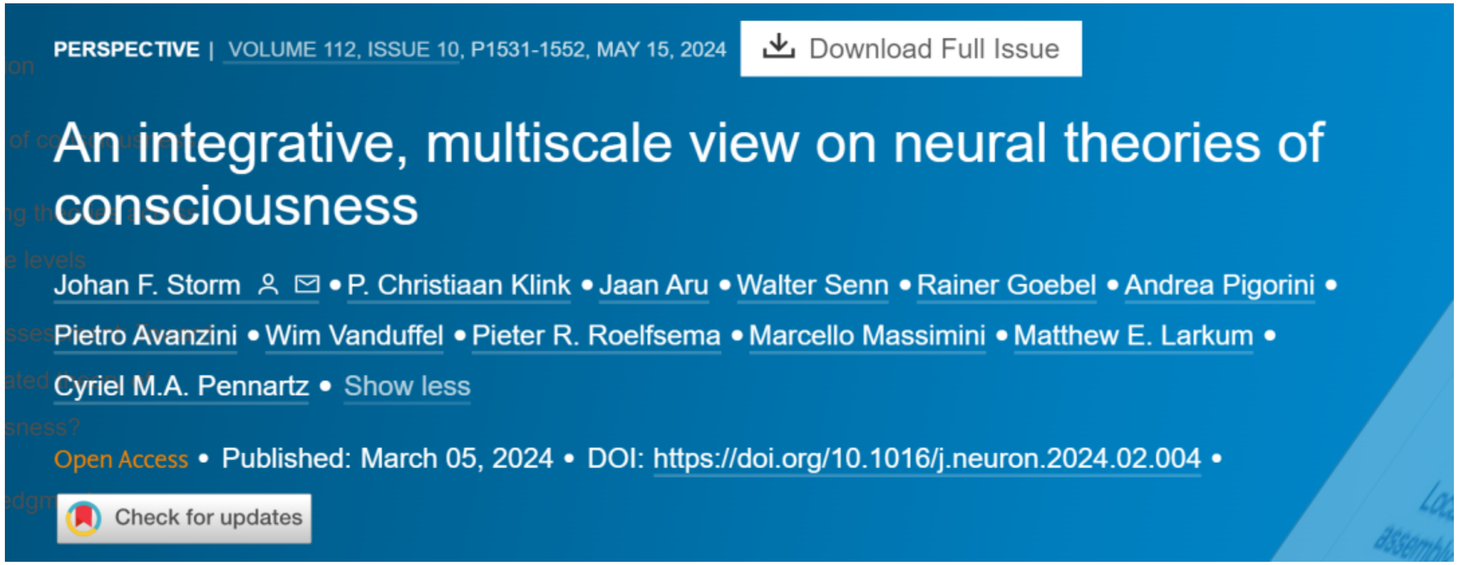
**Cogitate Consortium, Oscar Ferrante[1§], Urszula Gorska-Klimowska[2§], Simon Henin[3§], Rony Hirschhorn[4§], Aya Khalaf[5§], Alex Lepauvre[6,7§], Ling Liu[8,9§], David Richter[7,10,11§], Yamil Vidal[7§]**, Niccolò Bonacchi[12,13], Tanya Brown[6], Praveen Sripad[6], Marcelo Armendariz[14,15], Katarina Bendtz[14,15], Tara Ghafari[1], Dorottya Hetenyi[1,16], Jay Jeschke[3], Csaba Kozma[2,17], David R. Mazumder[14], Stephanie Montenegro[3], Alia Seedat[3], Abdelrahman Sharafeldin[18], Shujun Yang[19], Sylvain Baillet[20], David J. Chalmers[21], Radoslaw M. Cichy[22,23,24], Francis Fallon[25], Theofanis I. Panagiotaropoulos[26], Hal Blumenfeld[5], Floris P de Lange[7], Sasha Devore[3], Ole Jensen[1], Gabriel Kreiman[14,15], Huan Luo[8], Melanie Boly[2,27], Stanislas Dehaene[26,28], Christof Koch[29], Giulio Tononi[2], Michael Pitts*[30], Liad Mudrik*[4,31], Lucia Melloni*[3,6]




**Extended Data Figure 10: An overview of theoretical predictions, experimental outcomes and interpretations.**

# 意识理论的未来?

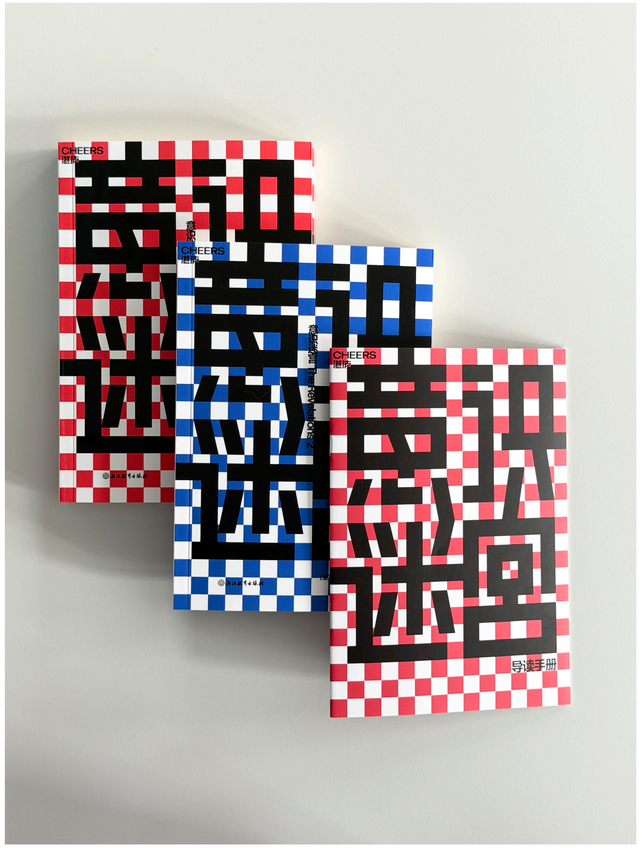
## 尝试整合现有的意识理论

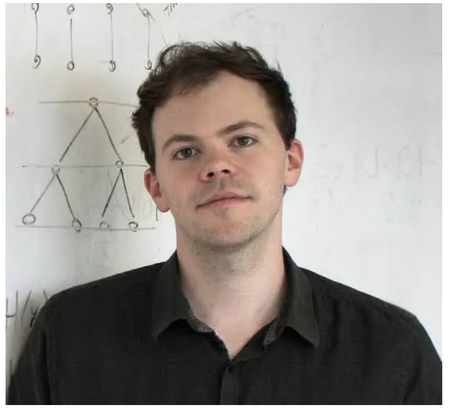


https://swarma.org/?p=51193

在一些人看来，好像我们对意识已经完全了解，甚至可以动手计算了；在另一些人看来，似乎什么都没说——「意识乃是至高至玄之物，非我等凡人可窥之堂奥」。但我恰恰就是要反对这两种观点：我们没有完美的意识理论，但也不是完全不能下手。

—章彦博



Erik Hoel

神经科学家

科幻小说作家

福布斯30位30岁以下科学家奖

因果涌现理论提出者

IIT理论贡献者

集智俱乐部

- 一部以意识科学研究领域为背景的谋杀悬疑小说。
- 这部兼具硬核科学底色与文学诗性的小说，以神经科学界的“圣杯”——意识本质研究为舞台，编织了一场横跨认知科学、存在主义哲学与技术伦理的多维狂想